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ExaAM: Metal additive manufacturing simulation at the fidelity of the microstructure

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Abstract

Additive manufacturing (AM), or 3D printing, of metals is transforming the fabrication of components, in part by dramatically expanding the design space, allowing optimization of shape and topology. However, although the physical processes involved in AM are similar to those of welding, a field with decades of experimental, modeling, simulation, and characterization experience, qualification of AM parts remains a challenge. The availability of exascale computational systems, particularly when combined with data-driven approaches such as machine learning, enables topology and shape optimization as well as accelerated qualification by providing process-aware, locally accurate microstructure and mechanical property models. We describe the physics components comprising the Exascale Additive Manufacturing simulation environment and report progress using highly resolved melt pool simulations to inform part-scale finite element thermomechanics simulations, drive microstructure evolution, and determine constitutive mechanical property relationships based on those microstructures using polycrystal plasticity. We report on implementation of these components for exascale computing architectures, as well as the multi-stage simulation workflow that provides a unique high-fidelity model of process–structure–property relationships for AM parts. In addition, we discuss verification and validation through collaboration with efforts such as AM-Bench, a set of benchmark test problems under development by a team led by the National Institute of Standards and Technology.

Keywords

3D printing, coupled physics, metal additive manufacturing, microstructure, multiscale materials

Introduction

Additive manufacturing (AM) is revolutionizing manufacturing, allowing construction of complex parts not readily fabricated by traditional techniques. In addition, AM offers the possibility of constructing "designer materials" by adjusting process control variables to achieve spatially varying physical properties. However, harnessing these opportunities for producing unique materials and components requires a detailed understanding of the thermal cycles present during AM processes and their effect on material structure and properties. Although there has been significant interest and investment in AM, the fraction of this investment devoted to modeling and simulation is relatively small and not focused on development of high-fidelity predictive models but instead on reduced-order models for industry use. Introducing high-fidelity predictive models into the design cycle will enable parts designed for AM processing,

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parts designed with location-specific material properties, and part qualification augmented by modeling and simulation.

In AM, a geometric description of the part is processed into 2D slices. A feedstock material is melted, and the part is built layer-by-layer. In metal AM, the feedstock is often in wire or powder form, and the energy source is a laser or electron beam. In powder bed processes, each layer is nominally 30–50 microns, so a part 1 cm tall would require 200 layers or more, each consisting of new feedstock powder and one or more passes of the laser or electron beam to sinter and/or melt the powder in specific locations.

A complex interplay among multiple physical phenomena at spatial and temporal scales spanning many orders of magnitude determines the performance of the final manufactured part (Figure 1). These include heat transfer (conductive, convective, radiative, and evaporative), fluid flow, melting and solidification, and solid-solid phase transformations. These phenomena are directly influenced by controllable process parameters, such as the pattern by which each layer is melted, the diameter, magnitude, and speed of the energy source. In turn, these phenomena influence the microstructure throughout the part, determining local properties, residual stress, and ultimately performance (e.g., strength, modal properties, and service life). This sequence is often referred to as the process–structure–property–performance (PSPP) relationship (DebRoy et al., 2018; Smith et al., 2016). There are gaps in understanding the integrated sequence for AM processes; filling these gaps where possible, and quantifying uncertainties where it is not, is key to unlocking the potential of AM.

The physical processes involved in AM are similar to those in welding, a field with a wealth of experimental, modeling, simulation, and characterization research over the past decades. Unfortunately, the simulation tools developed for welding and other similar processes, while calibrated and approaching predictive capability, are inadequate for AM processes, as demonstrated by the inability to predict the build failure rate for new AM parts, which often exceeds 20% (Yavari et al., 2019).[1] We believe this is largely due to the fact that PSPP relationships are traditionally analyzed in an uncoupled manner, relying on tabular databases unable to adequately capture the implicit, dynamic, non-equilibrium nature of AM processes. Exascale computational resources provide the opportunity to remove some of these limitations by coupling high-fidelity mesoscale simulations within continuum process simulations to determine microstructure and properties using local conditions.

The U.S. Department of Energy's Exascale Computing Project (ECP)[2] is preparing for the arrival of exascale systems such as Frontier[3] at the Oak Ridge Leadership Computing Facility (OLCF)[4] at Oak Ridge National Laboratory by funding accelerated hardware roadmaps as well as a software ecosystem and a broad range of applications (Alexander et al., 2020). As part of ECP, the Exascale Additive Manufacturing Project (ExaAM) is developing a multi-stage simulation workflow that provides a unique high-fidelity model of process–structure–property relationships for AM parts. We are using pre-exascale platforms such as Summit[5] (also at OLCF) and Sierra[6] (at Lawrence Livermore National Laboratory[7]) for development, performance analysis, and optimization. Summit is an IBM AC922 system with 4608 compute nodes, each of which has two IBM POWER9TM CPUs (168 total hardware threads), six NVIDIA VoltaTM V100 GPUs and 608 GB of memory (512 GB DDR4[8] and 96 GB HBM2[9]). The nodes are connected by a Mellanox 100 Gb/s EDR Infiniband interconnect,[10] and the aggregate system has a peak performance of 200×10^{15} floating point operations per second (200 petaflops).

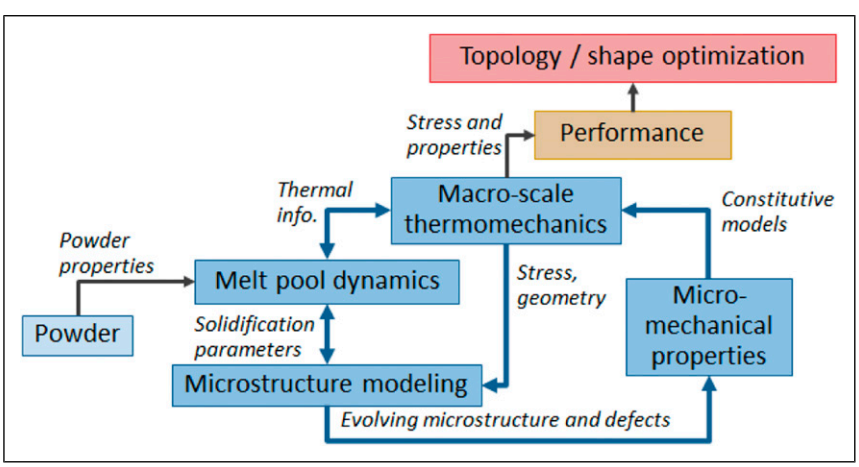

Figure 1. Coupled multiscale physical phenomena inherent in metal additive manufacturing.

Challenge problem description

Comprehensive AM simulations at the fidelity of the microstructure must span numerous orders of magnitude in both length and time while incorporating a broad range of coupled physical phenomena. Measurements for guiding and validating these simulations must be similarly broad with all the relevant measurements coordinated to provide a quantitative description of a complete AM process, from characterization of the feedstock material, through *in situ* monitoring of the build process, to comprehensive characterization of the as-built material residual stresses and position-dependent microstructure. The Additive Manufacturing Benchmark Test Series (AM-Bench)[11] was developed by the National Institute of Standards and Technology (NIST) to provide such data for validating and guiding AM simulations (Levine et al., 2020). ExaAM has adopted the AM-Bench measurements to define the relevant challenge problem, test its simulation results, and enable cross comparisons with simulations from other groups around the world (Anderson et al., 2019; Fan and Li, 2019; Gan et al., 2019; Kollmannsberger et al., 2019; Megahed et al., 2019; Robichaud et al., 2019; Yang et al., 2019).

The set of measurements adopted for the ExaAM Challenge Problem are the 2018 AM-Bench 3D builds and measurements (AMB2018-01) of the nominally single-phase, nickel-based superalloy known as Inconel 625 (IN625). Figure 2(a) shows an example build plate of AMB2018-01 test parts and Figure 2(b) shows the nominal part dimensions. The part uses a bridge design with legs of different thicknesses (0.5 mm, 2.5 mm, and 5.0 mm) to explore the effect of wall thickness on cooling rate, melt pool length, and microstructure evolution. The laser power was 195 W with a scan speed of 800 mm/s and a hatch spacing of 100 μm. The laser diameter at the build plane is 85 μm during the contour scans and 100 μm during the infill scans. The infill scan pattern is parallel to the X axis (along the length of the part) during odd-numbered layers and parallel to the Y axis during even-numbered layers. The build platform is lowered 0.02 mm between each layer to spread a new layer of virgin powder. The publicly available AMB2018-01 data sets provide feedstock material characterization,[12] a detailed description of the spot and laser scan pattern (Heigel et al., 2020b), *in situ* thermocouple measurements within the build chamber (Heigel et al., 2020b), time resolved measurements of cooling rate and melt pool length (Heigel et al., 2020a, 2020b) within each layer in the field of view as shown in Figure 2(b)), *ex situ* characterization of the residual strains and stresses using synchrotron X-rays, neutron diffraction, and the contour method (Phan et al., 2019), part distortion measurements after cutting part of the specimen off the build plate (Phan et al., 2019), multiple laboratory and synchrotron X-ray microstructure studies as a function of position and orientation within the build (Stoudt et al., 2020; Zhang et al., 2019a), and measurements of the microstructure evolution during a residual stress relieving heat treatment (Zhang et al., 2019a). The builds and measurements for AMB2018-01 were conducted jointly by 21 researchers from six different organizations (Levine et al., 2020). Additional measurements focusing on the mesoscale and macroscale mechanical properties of the AMB2018-01 parts and the microstructure of larger areas are in progress.

Several of the ExaAM simulations require additional validation data at the critical melt pool level and these data are provided through the AMB2018-02 AM-Bench measurements. Single laser tracks with different laser power and speed combinations were conducted on bare metal plates of IN625 with time resolved measurements of melt pool length and cooling rate and post-solidification cross-sectional measurements of the melt pool transverse geometry (Lane et al., 2020a). Topographic measurements of the 3D geometry of the solidified melt tracks were conducted (Ricker et al., 2019) along with microstructure characterization of the cross sections (Stoudt et al., 2020). The measurements for AMB2018-02 were conducted jointly by

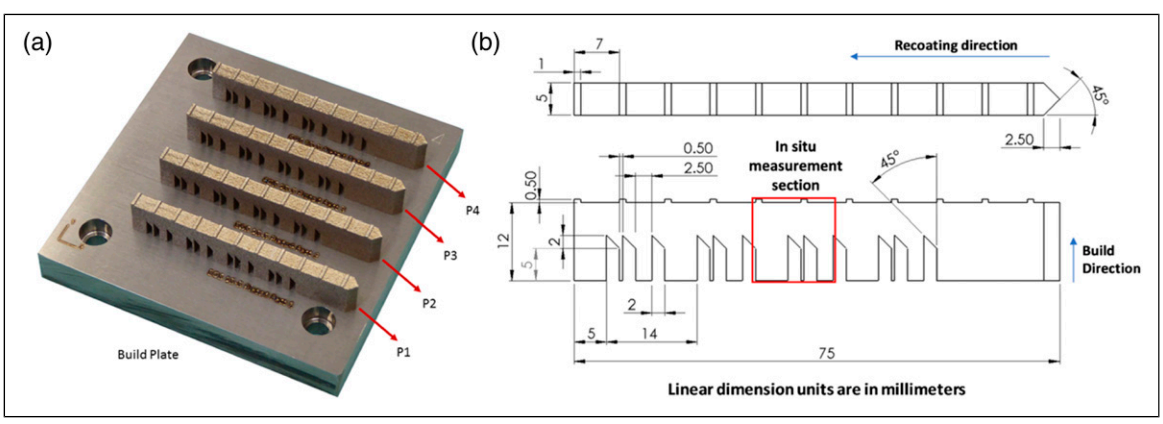

Figure 2. AMB2018-01.

14 researchers from two organizations (Levine et al., 2020). Additional measurements of *in situ* laser absorptivity (Lane et al., 2020b) and energy balance are in progress.

Computational workflow

We have decomposed the coupled problem depicted in Figure 1 into five stages, outlined in Figure 3, with more detail shown in the Unified Modeling Language[13] Activity Diagram in Figure 4. A critical observation is that there is no single code able to capture all of the relevant physics required; indeed, different numerical approaches are best suited for the AM sub-problems. However, several codes have been developed to simulate phenomena similar to those required for AM, both within the Department of Energy complex and the broader computational science community. ExaAM is leveraging several of those existing capabilities—enhancing and extending them as needed—and developing new capabilities where necessary.

Typically, thermo-mechanical finite element models are employed at the macroscopic part scale; finite volume or finite element models for fluid dynamics and heat transfer to capture the melt pool dynamics and solidification at millimeter scales; mesoscale approaches (e.g., discrete elements, cellular automata, kinetic Monte Carlo, or phase-field models) to simulate melting, solidification, and microstructure formation at the micrometer scale; and polycrystal plasticity models to develop the microscale mechanical property relationships. The components being developed in ExaAM are summarized in Tables 1 and 2. Stages 0 and 2 to 4 are all performed by single components as shown in Table 1. Stage 1 encompasses enough complexity that multiple components are required, as indicated in Table 2. The primary components are TruchasPBF and OpenFOAM for melt pool simulation (sometimes referred to as Stage 1a) and ExaCA for microstructure evolution (Stage 1b). Additionally, ExaMPM represents a new and promising approach to accurately resolve the multi-phase dynamics of solid, liquid, vapor, and powder, as well as laser-matter interactions. Two phase-field components, Tusas and AMPE, inform parameters used by ExaCA to describe dendrite growth kinetics during solidification (Table 3).

Stage 0: Initial full part build

As discussed above, our goal is to utilize exascale platforms to simulate AM processing at the fidelity of the microstructure as required to determine process-aware material properties and, ultimately, the performance of an AM built part. Diablo is a parallel thermomechanics code that uses a continuum representation of the local material behavior which may be unique to AM built materials. In this section, we describe Diablo capabilities that have been implemented for simulation of AM build processes on exascale architectures, including adaptive mesh refinement (AMR) and parallel-in-time parallelization.

The initial full part simulation is a finite element solution of coupled heat transfer and solid mechanics, as adapted to the laser powder bed fusion (LPBF) problem, with the ultimate goal of providing an initial thermal history and a prediction of the as-built configuration. The strong form of the heat transfer problem is given by

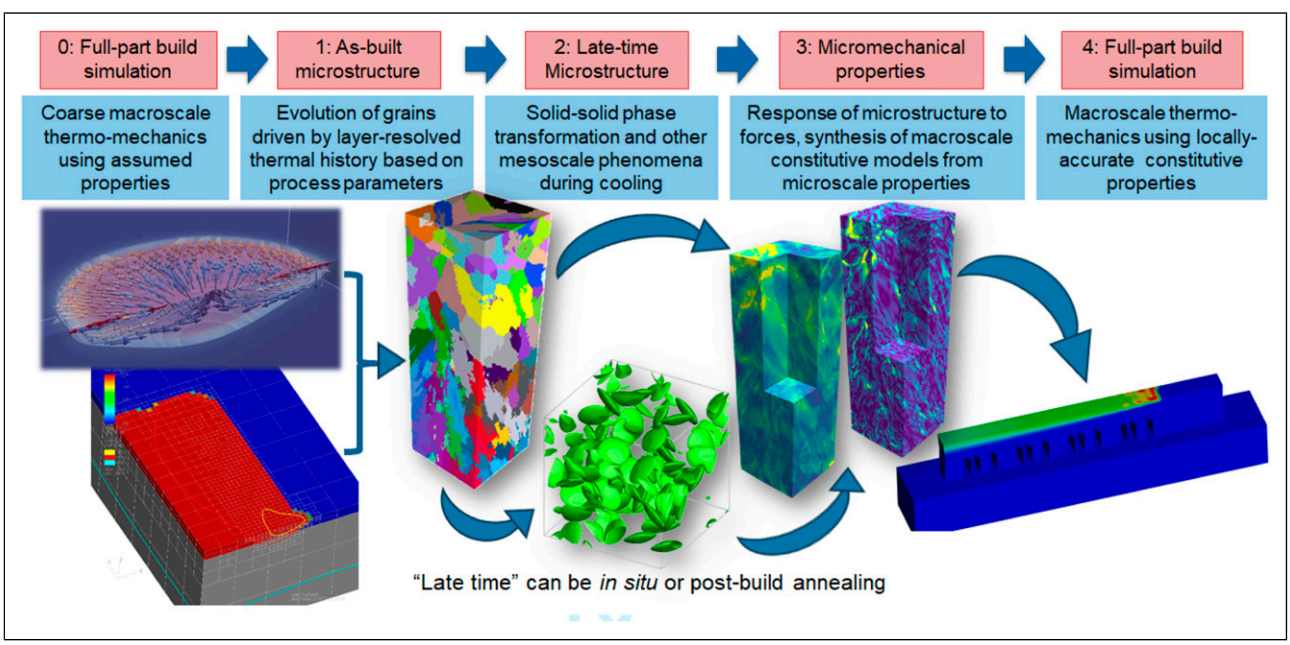

Figure 3. The five stages of the ExaAM simulation workflow.

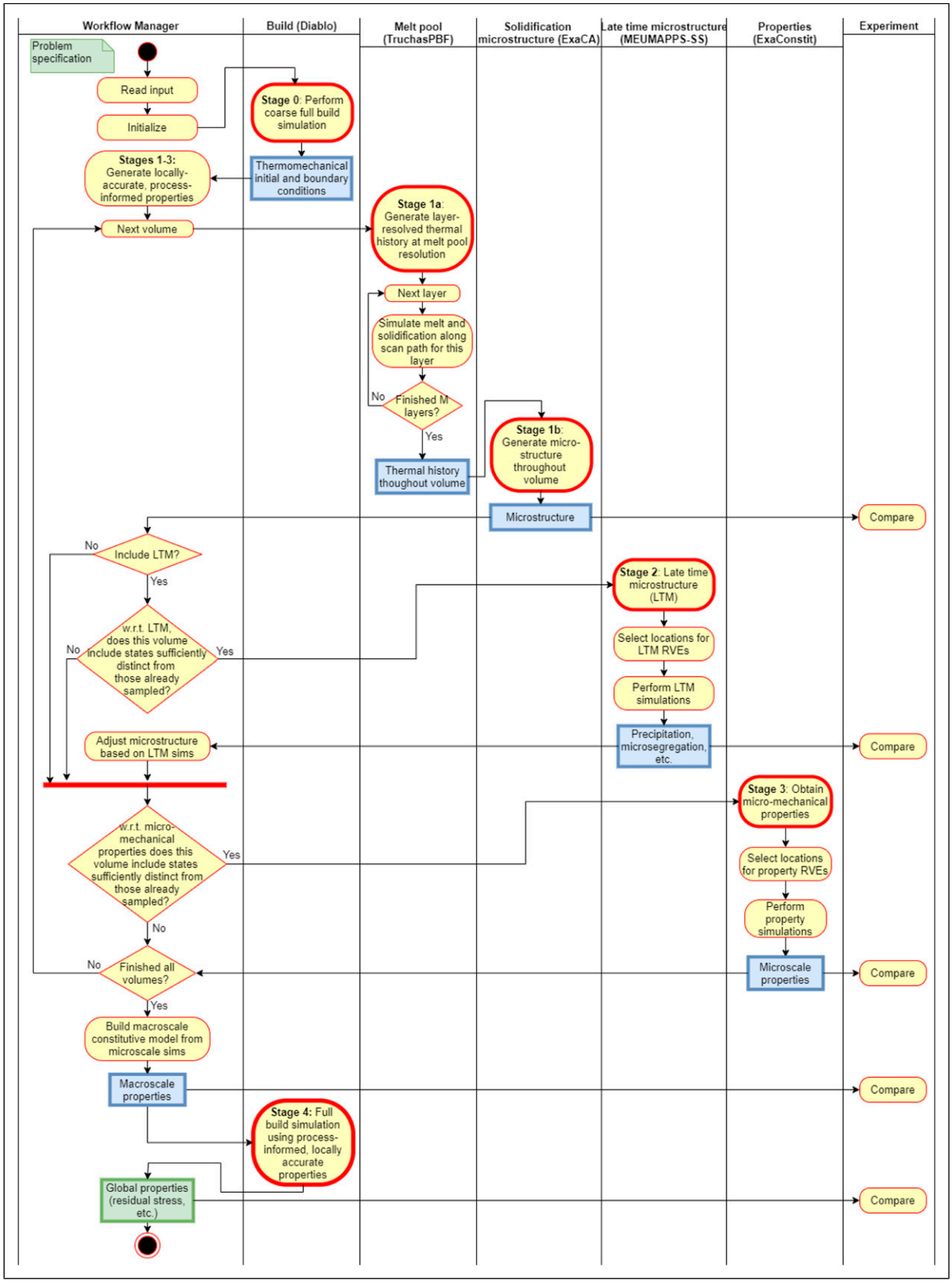

Figure 4. Unified Modeling Language[13] activity diagram of the ExaAM (Exascale Additive Manufacturing) simulation workflow. Note that RVE is "representative volume element".

Table 1. Components implementing Stages 0, 2, 3, and 4 of the ExaAM Simulation Workflow (Figures 3 and 4).

Stage(s)	Code	Physical/numerical models	Computational motifs
0,4: Full build simulation	Diablo	Implicit solid mechanics, heat and mass transfer, contact	Lagrangian FEM, unstructured grids, adaptive h-refinement, nonlinear physics
2: Late-time microstructure	MEUMAPPS-SS	Phase-field, solid–solid phase transformation	Fourier spectral method, structured grids
3: Micromechanical properties	ExaConstit	Polycrystal plasticity	Lagrangian FEM, unstructured grids, nonlinear mechanics

Table 2. Components implementing Stage 1 of the ExaAM Simulation Workflow (Figures 3 and 4).

Sub-stage	Code	Physical/numerical models	Computational motifs
Melt pool dynamics	TruchasPBF	Free-surface flow, heat transfer, phase change, species diffusion	Eulerian FVM, structured grids, AMR, implicit mimetic finite difference, sparse linear algebra, nonlinear solvers
	OpenFOAM	Fluid flow, heat transfer, phase change	Eulerian FVM, structured grids, dynamic time stepping, adaptive time integration
Microstructure evolution	ExaCA	Cellular automata, grain growth	FDM, structured grids
Sub-grain microstructure	Tusas	Phase-field, solidification	Implicit FEM, preconditioned Jacobian-free Newton-Krylov (JFNK), unstructured 2D and 3D
	AMPE		Implicit FVM, linear and nonlinear solvers, structured AMR
Powder-resolved melt pool dynamics	ExaMPM	Free-surface flow, heat transfer, phase change	Material point method, structured grids, sparse linear algebra, particle methods

Table 3. Where to look for more information about ExaAM components.

Code	Reference(s)
Diablo	Hodge et al. (2014)
TruchasPBF	https://www.truchas.org/
OpenFOAM	https://www.openfoam.com/
ExaCA	Gandin and Rappaz (2020), https://github.com/LLNL/ExaCA
Tusas	Trainer et al. (2016), https://github.com/chrisknewman/tusas
AMPE	http:://github.com/LLNL/AMPE
ExaMPM	https://github.com/ECP-copa/ExaMPM
MEUMAPPS-SS	Radhakrishnan et al. (2016), https://github.com/ORNL/meumapps_ss
ExaConstit	Carson et al. (2019), https://github.com/LLNL/ExaConstit

$$\rho c_p = -(\nabla \cdot \mathbf{q}) + r, \quad \dot{T} \text{ in } \Omega \tag{1a}$$

$$T(\mathbf{x}_T,t) = \overline{T}, \quad \text{on } \mathbf{x}_T \in \Gamma_T \tag{1b}$$

$$q(\mathbf{x}_q,t) = \overline{\mathbf{q}} \cdot \mathbf{n}, \quad \text{on } \mathbf{x}_q \in \Gamma_q \tag{1c}$$

$$T(\mathbf{x},0) = T_0, \quad \text{on } \Omega \bigcup \partial\Omega \tag{1d}$$

where T is the temperature, and the flux $\mathbf{q}$ is represented by the typical Fourier conduction

$$\mathbf{q} = -\mathbf{k}\nabla T \tag{2}$$

Additionally, $\overline{T}$ is the value of the Dirichlet boundary condition, is the value of the Neumann boundary condition, and T_0 is the value of the initial condition. The parameters ρ, c_p, $\mathbf{k}$, and $\overline{\mathbf{q}}$ are, in general, all functions of the temperature T. An additional equation is solved for the interface problem representing phase change, with state variables ϕ_1 and ϕ_2 used to track powder and consolidated materials, respectively, and the volumetric load term r represents the laser as a moving heat source. The solid mechanics problem (consisting of balance laws for mass and linear momentum) is given by

$$\dot{\rho} = \rho \nabla \cdot \mathbf{v}, \quad \text{in } \Omega \tag{3a}$$

$$\rho \dot{\mathbf{v}} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}, \quad \text{in } \Omega \tag{3b}$$

$$\mathbf{u}(\mathbf{x}_u,t) = \overline{\mathbf{u}}, \quad \text{on } \mathbf{x}_u \in \Gamma_u \tag{3c}$$

$$\mathbf{t}(\mathbf{x}_t,t) = \overline{\mathbf{t}}, \quad \text{on } \mathbf{x}_t \in \Gamma_t \tag{3d}$$

$$\rho(\mathbf{x},0) = \rho_0, \quad \text{on } \Omega \bigcup \partial\Omega \tag{3e}$$

$$\mathbf{u}(\mathbf{x},0) = \mathbf{u}_0, \quad \text{on } \Omega \bigcup \partial\Omega \tag{3f}$$

$$\mathbf{t}(\mathbf{x},0) = \mathbf{t}_0, \quad \text{on } \Omega \bigcup \partial\Omega \tag{3g}$$

where $\mathbf{u}$ is the displacement, the tensor $\boldsymbol{T}$ is the Cauchy stress (as opposed to the scalar T associated with the heat

transfer problem), Γ_u is the portion of the boundary $\partial\Omega$ associated with essential boundary conditions, and Γ_t is the portion of the boundary associated with natural boundary conditions. The balance of angular momentum is satisfied by the constraint

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T \tag{4}$$

The coupling between the thermal and solid mechanical problems occurs through several mechanisms. Indeed, the Cauchy stress $\boldsymbol{\sigma}$ is not just a function of deformation $\mathbf{u}$, rather

$$\boldsymbol{\sigma} = \widehat{\boldsymbol{\sigma}}(\nabla\mathbf{u}, T, \ldots) \tag{5}$$

Moreover, this functional dependence occurs in two manners. First, the material parameters that define the stress response are a function of temperature, phase, and other microstructural descriptors (e.g., Young's modulus $E = \widehat{E}(T, \varphi_1, \varphi_2)$). Second, the strain contains constitutively defined parts that are functionally dependent on the temperature and the phases. The final coupling mechanism is that the solid mechanics problem moves the mesh, which changes the problem domain for the thermal problem. After weakening of the weighted residual associated with the balance laws, and applying a finite element approximation with the approximation of the temperature as

$$T = \sum_A N_A \widehat{T}_A \tag{6}$$

the resulting system of discrete equations for, say, the heat transfer problem, evaluated at time $n + 1$ is

$$\mathbf{M}(T_{n+1})\dot{\widehat{\mathbf{T}}}_{n+1} + \mathbf{K}(T_{n+1})\widehat{\mathbf{T}}_{n+1} = \mathbf{F}_{n+1} \tag{7}$$

where $\widehat{\mathbf{T}}$ represents the vector of nodal temperatures. The nonlinear nature of the problem requires an iterative solution, and the current case uses the Newton–Raphson method, which for the heat transfer problem, can be written as

$$\widehat{\mathbf{T}}_{n+1}^{(k+1)} = \widehat{\mathbf{T}}_{n+1}^{(k)} + \Delta\widehat{\mathbf{T}}_{n+1}^{(k)} \tag{8}$$

where the notation $(\cdot)^{(k)}$ indicates the kth Newton–Raphson iteration. If we define the residual $\mathbf{R}$ by taking all of the terms in (7) to one side of the equation and setting them equal to zero, then the increment in the solution can be defined by setting the linear portion of the residual $\mathbf{R}$ in the direction of the increment itself equal to zero, which results in

$$\Delta\widehat{\mathbf{T}}_{n+1}^{(k)} = -\left[\mathrm{D}_{\widehat{\mathbf{T}}}\mathbf{R}\left(\mathbf{T}_{n+1}^{(k)}\right)\right]^{-1}\mathbf{R}\left(\widehat{\mathbf{T}}_{n+1}^{(k)}\right) \tag{9}$$

where $\mathrm{D}_{\widehat{\mathbf{T}}}\mathbf{R}$ is taken to be the Fréchet derivative of $\mathbf{R}$ with respect to $\widehat{\mathbf{T}}_{n+1}$; Fréchet derivatives are described at length in various sources. Note that the terms on the left side of (7) can be differentiated using the product rule and the chain rule; the only term that may not be obvious is $\mathrm{D}_{\widehat{\mathbf{T}}}\widehat{\mathbf{T}}_{n+1}$. Given that we will perform serial time stepping based on the generalized trapezoidal time integrator (Hughes, 1987)

$$\widehat{\mathbf{T}}_{n+1} = \widehat{\mathbf{T}}_n + dt\left((1-\alpha)\dot{\widehat{\mathbf{T}}}_n + \alpha\dot{\widehat{\mathbf{T}}}_{n+1}\right) \tag{10}$$

we can calculate an algorithmic derivative as

$$\begin{aligned}\mathrm{D}_{\widehat{\mathbf{T}}}\dot{\widehat{\mathbf{T}}}_{n+1} &= \mathrm{D}_{\widehat{\mathbf{T}}}\left(\frac{\widehat{\mathbf{T}}_{n+1} - \widehat{\mathbf{T}}_n - (1-\alpha)dt\dot{\widehat{\mathbf{T}}}_n}{\alpha dt}\right) \\ &= \frac{1}{\alpha dt}\mathbf{I}\end{aligned} \tag{11}$$

for the case of $\alpha > 0$. As such, we can calculate the increment in the solution associated with the current nonlinear iteration, and terminate the iterative process when the choice of criteria is satisfied.

The description to this point is basically that described in Hodge et al. (2014) and Hodge et al. (2016). However, in the intervening time, improvements have been made to the modeling strategies used for this portion of the overall problem. The first of these is with respect to the constitutive model. In particular, the original rate independent plasticity was replaced by a model where the underlying form of the material behavior is itself a function of temperature, and can be briefly summarized as follows:

1. For $T < 0.5T_m$, strain rate independent plasticity (similar to the original formulation);
2. For $0.5T_m < T < 0.8T_m$, viscoplasticity, with the plastic portion being rate dependent; and
3. For $0.8T_m < T$, Maxwell-type viscoelasticity,

where T_m is the melting temperature, and with appropriate mechanisms included to ensure stress continuity across the temperature boundaries. This model was implemented in response to a better understanding of various experimentally observed behaviors, as well as to allow the modeling of annealing, both *in situ* and post-build. A complete exposition on this material model can be read in Ganeriwala et al. (2019).

The next significant improvement relates to the representation of the multiple spatial scales present in the LPBF problem. Note that a cubic domain, measuring 2 cm on an edge, with 30 μm layers requires a minimum of around 300 million elements for a uniform hexahedral mesh. In order to reduce the meshing requirement, the Diablo developers extended the native, fully parallelized h-refinement implementation to support the LPBF problem. In particular, several features were added, including the ability to refine based on the surface growth of the material and to properly

handle the implied discontinuities induced in the reference configuration (e.g., Figure 5). Several additional refinement criteria were also implemented, including the ability to resolve gradients between the powder and consolidated material properties (as depicted in Figure 6), as well as the ability to have a region of mesh refinement track the moving heat source. An important implication of this capability is that it allows users to represent only the desired regions with higher mesh resolution, and the remainder of the domain may be represented with as coarse a mesh as one can tolerate. Care was taken to ensure a consistent implementation, in that reductions in accuracy are in agreement with analytical predictions. A full description of the updated AMR capabilities is contained in the work of Ganeriwala et al. (2021).

The final improvement being implemented addresses the temporal scales of the problem. Note that the cubic domain described above, with 60 μm track spacing, a typical serpentine path with a heat source traveling at 600 mm/s, and the material nonlinearities described above, requires between 131 million and 874 million time steps. As such, a more sophisticated representation than that allowed by uniform serial time stepping is needed. To this end, multirate time integration (MTI) and parareal-style parallel-in-time (PIT) are being implemented, simple examples of which are depicted by Figures 7 and 8, respectively. Conservatively estimated performance improvements[14] are 90 times speedup for MTI and 40 times speedup for PIT, if used individually.[15] Note that the conservatism of the PIT estimate comes from the fact that a *single layer* of the 2 cm cube mentioned above would benefit from access to nearly 900 temporal subintervals (i.e., 900 copies of the underlying spatial problem), yet testing has currently only been performed on around 1/3 of that; if the hardware resources were available, we estimate that the speedup would be between 100 and 120 times for PIT. Additionally, the trial implementations of both MTI and PIT are exhibiting the expected convergence properties. A full

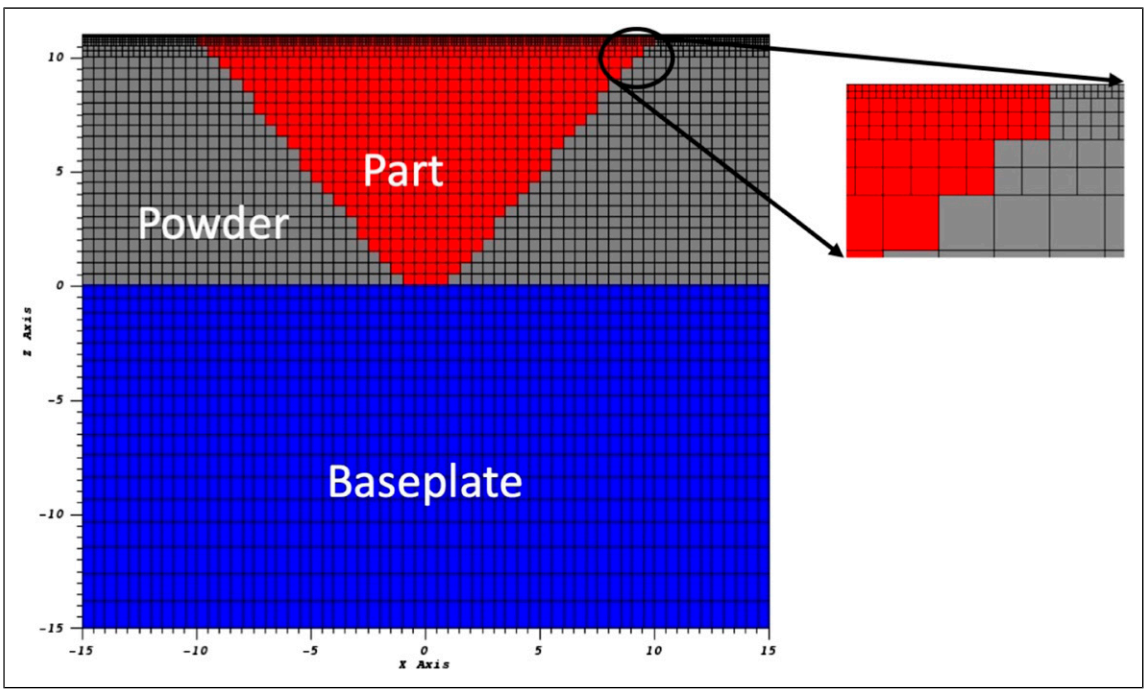

Figure 5. Example of layerwise mesh refinement for the part-scale problem (Stage 0).

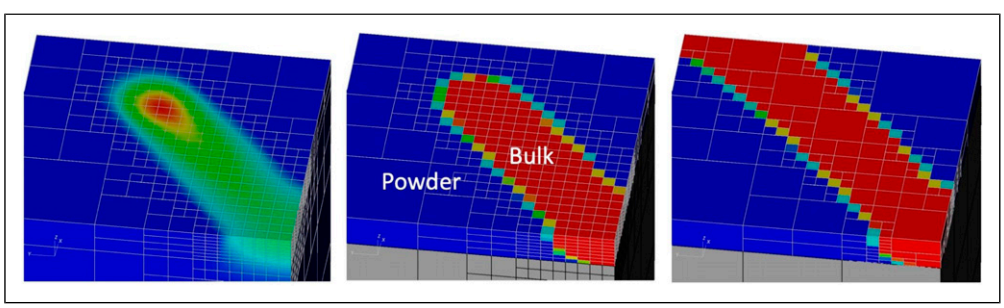

Figure 6. Example of representation of phase gradients at homogeneous temperatures for the part-scale problem (Stage 0).

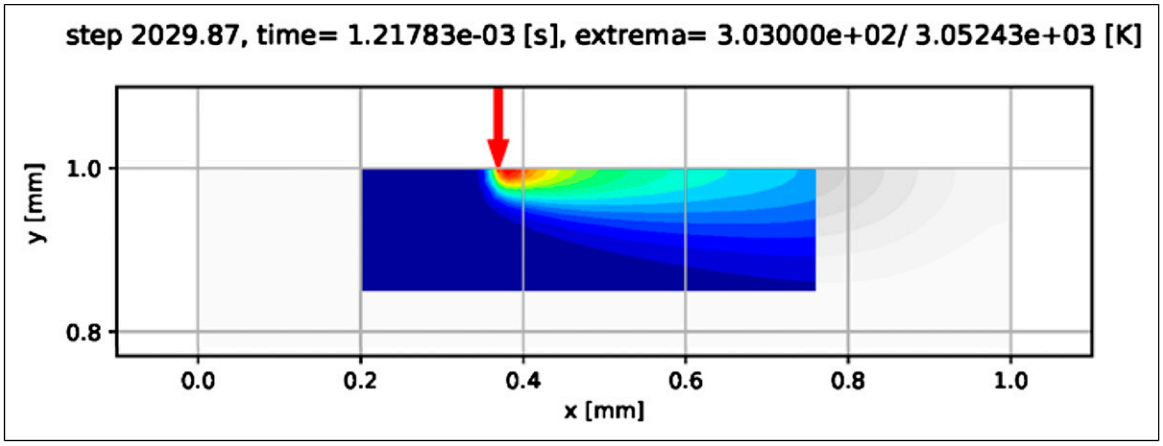

Figure 7. Typical thermal contours of a multirate time integration solution for the part-scale problem (Stage 0).

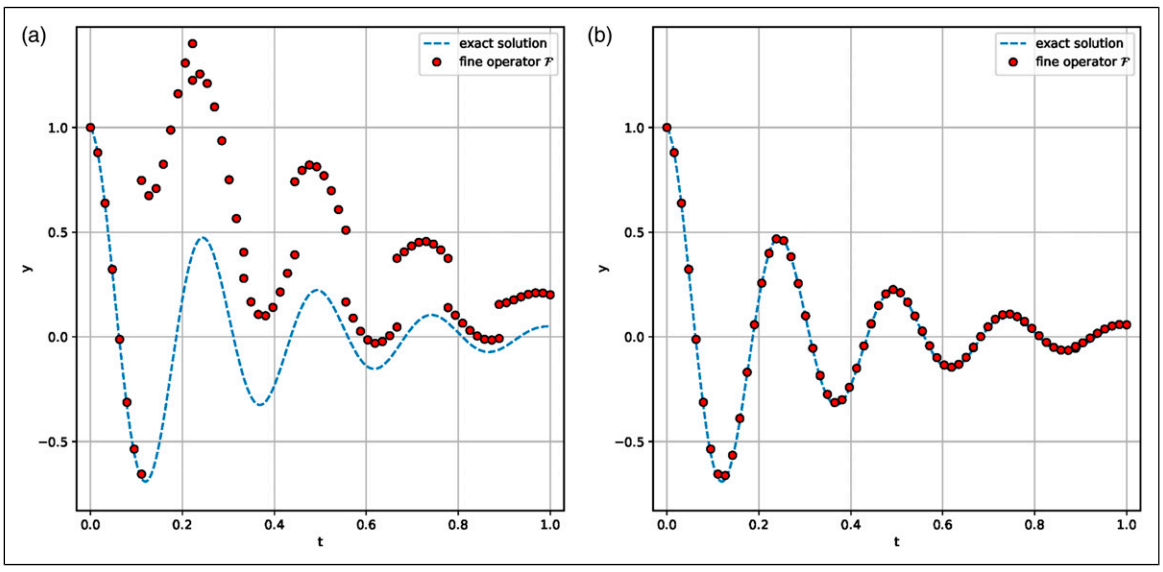

Figure 8. Example time history of a simple scalar function calculated using parallel-in-time for the part-scale problem (Stage 0): (a) Solution after first iteration and (b) converged solution.

description of the improved time evolution methods is presented in the work of Hodge (2020).

Stage 1: As-built microstructure

Melt pool (1a). The Diablo simulation, while an accurate model of the thermo-mechanical state at every location, does not resolve the melt pool as needed for the solidification simulation of microstructure development. As such, we have created an AM powder bed fusion version of the Truchas continuum solidification code (TruchasPBF) to compute the melt pool scale solidification within a given thermo-mechanical state (Figure 3). While TruchasPBF is designed for exascale architectures, AM processing is an inherently sequential process and simulating a location (20 layers) sequentially at the scale of the melt pool is beyond the capability of even exascale systems. To solve this problem and more fully utilize exascale platforms, we use a parallel-in-time algorithm to divide a scan pattern into hundreds of segments which are then solved iteratively.

Modeling the melting of the powder and substrate, fluid flow within the melt pool and the subsequent resolidification is an extremely challenging problem. High-fidelity, computationally expensive models have been demonstrated for very short portions of laser track (Khairallah and Anderson, 2014; Khairallah et al., 2016). However, within the context of full component simulation, we have employed a simplified model of melt pool behavior through the use of the open source TruchasPBF code. TruchasPBF is a finite volume thermal-fluid modeling code built on the AMReX framework (Zhang et al., 2019b). Its capabilities include incompressible, multi-phase, laminar fluid dynamics, flat-free surface of the molten alloy, heat transfer, phase change, evaporation, and surface tension phenomena. A representative large-scale melt pool

simulation showing streamlines at cross sections through the molten area of the melt pool is shown in Figure 9.

TruchasPBF solves the following conservation law for heat conduction and enthalpy advection

$$\frac{\partial H}{\partial t} + \nabla \cdot \mathbf{q} + Q_a = Q(\mathbf{x},t) \tag{12a}$$

$$H = f(T) \tag{12b}$$

$$q = -k\nabla T \tag{12c}$$

$$Q_a = \nabla \cdot H\mathbf{v} \tag{12d}$$

Here, H is the enthalpy density, T is the temperature, k is the conductivity, and Q is a volumetric source term due to the laser. Q_a is the enthalpy advection term, dependent on the velocity $\mathbf{v}$ provided by the fluid mechanics solver. These equations are integrated in time using the BDF2 method. Each time step's solution is solved using a nonlinear Krylov accelerated fixed point iteration, with a PFMG preconditioner.

For a given cell j, conductivity and enthalpy density are functions of temperature and composition (given by the volume fractions of liquid and solid, α^l and α^s). The liquid fraction, and enthalpy density and conductivity of each phase is computed from user-defined functions of temperature

$$\alpha_j^l = \alpha^l(T) \tag{13a}$$

$$\alpha_j^s = 1 - \alpha_j^l \tag{13b}$$

$$H_j = \alpha_j^l H^l(T) + \alpha_j^s H^s(T) \tag{13c}$$

$$k_j = \alpha_j^l k^l(T) + \alpha_j^s k^s(T) \tag{13d}$$

TruchasPBF offers as boundary conditions Dirichlet temperature, user-provided heat flux, an external heat flux condition dependent on a user-provided coefficient and ambient temperature, heat flux contribution due to thermal radiation to an ambient external temperature, and heat flux contribution due to evaporation. In the context of additive manufacturing, a heat flux of zero (adiabatic) is enforced on the sides of the domain, except the top where evaporation and radiation terms are active. The laser may either be modeled as a heat flux boundary condition or a volumetric source term, in both cases as a user-defined function of time and space.

Within the melt pool, flow is modeled using the incompressible Navier–Stokes equations

$$\nabla \cdot \mathbf{v} = 0 \tag{14a}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot \rho \mathbf{v}\mathbf{v} = -\nabla p + \nabla \cdot \left(\mu\left(\nabla \mathbf{v} + \nabla^{\mathrm{T}}\mathbf{v}\right)\right) + \mathbf{f}_B + \mathbf{f}_S \tag{14b}$$

$$\mathbf{f}_B = \rho(T)\mathbf{b} \tag{14c}$$

$$\mathbf{f}_S = \delta_S \nabla_S \sigma \tag{14d}$$

Here, $\mathbf{v}$ is the velocity, p is pressure, ρ is density, $\mathbf{b}$ is an acceleration vector, σ is the surface tension coefficient, and μ is the dynamic viscosity. The buoyant body force $\mathbf{f}_B$ is computed using the Boussinesq approximation, such that the temperature-dependence of the fluid density appears only in that term. The normal component of surface tension is neglected due to the 0-curvature boundary, leaving only the tangential component as a traction at that boundary, here given as $\mathbf{f}_S$. This produces the Marangoni surface tension effect, which drives momentum and thermal circulation in the melt pool.

At liquid–solid interfaces, no-slip velocity and Neumann pressure boundary conditions are enforced. The top of the melt pool is a free surface modeled as a series of boundary

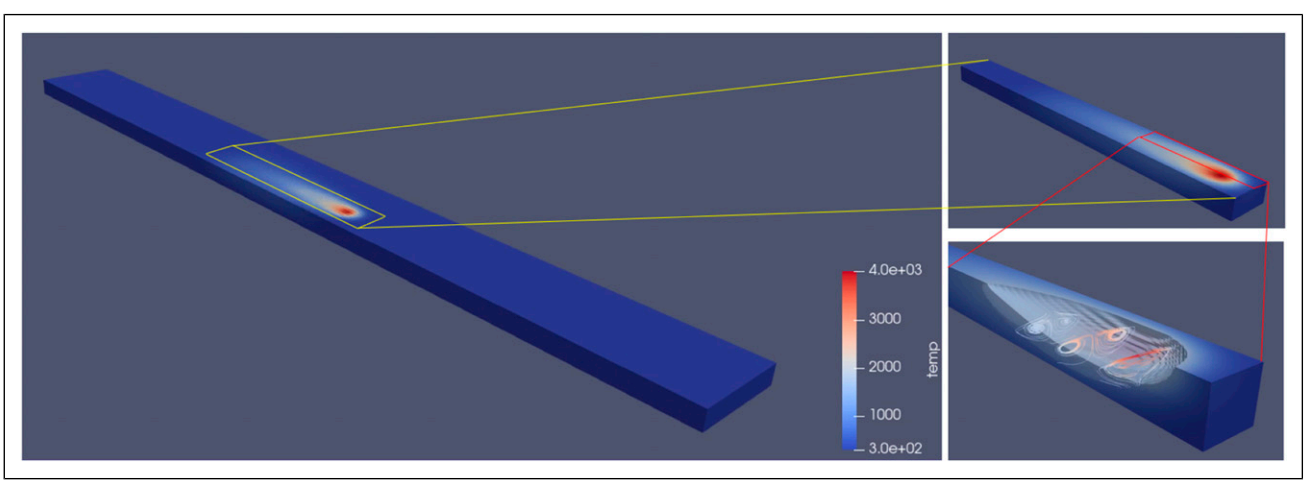

Figure 9. High-fidelity TruchasPBF simulation (Stage 1a).

conditions: free-slip velocity (no penetration, and 0-Neumann in the tangential directions), Dirichlet-0 pressure, and tangential surface tension.

The flow equations are integrated in time using the pressure projection method, with a semi-implicit viscous predictor step and a pressure Poisson system. Both systems are solved with HYPRE's hybrid CG solver (Falgout et al., 2006), which dynamically selects either diagonal-scaling or algebraic multigrid as a preconditioner depending on convergence rates.

As previously mentioned, one of the primary modeling challenges for an entire part is the drastically varying length and time scales. While the NIST AMB2018-01 challenge problem part is several centimeters in size, zone sizes of 5 microns are often needed to resolve the heat transfer and fluid flow in the melt pool. Such small zones, coupled with a relatively fast-moving heat source, generally require a very small time step, typically on the order of 1 μs or less, compared to the hours it takes to actually build the component. This large discrepancy in time scales means that billions of time steps are needed to simulate the manufacture of an entire component which is impractical for a typical multi-physics simulation code (MPI communication costs alone would be measured in days). To address this challenge, it is clear that the time domain must be parallelized. It is also important to note that high mesh resolution is only needed in the vicinity of the melt pool. This means that with an adaptive meshing strategy, the total number of zones that are needed in a simulation is actually comparatively small (often less than 500k), which means that strong scaling is typically limited to less than 50 MPI ranks.

Given initial conditions for each interval, parallelizing the time domain into independent interval simulations is trivial. Recognizing that the strongest nonlinearities occur only in the immediate vicinity of the melt pool, we developed an iterative technique that relies on the principle of superposition to accurately estimate the initial conditions for a large number of independent interval simulations. Each interval simulation is small since the mesh only needs to be refined in the vicinity of laser activity for that interval. For the NIST AMB2018-01 challenge problem, each interval easily fits on a single node of Summit,[5] and we have a computationally scalable approach to simulating melt pool behavior for an entire build.

Given the definitions in Table 4, Algorithm 1 describes the process for obtaining temperature distributions at the start of each interval (see Bement et al. (2020) for details of the full procedure). Algorithm 5 has been applied to the NIST AMB2018-01 benchmark, and an example section is shown in 10a. In this example, the laser scan path has been broken into 25 intervals. As shown in Figure 10(b), the error in the temperature at the beginning of each of these intervals, relative to a monolithic single simulation of the entire laser scan path, decreases rapidly with only a couple iterations. Using this approach, high-fidelity thermal history modeling can be performed over many layers. This high-fidelity thermal history is then used as an input for modeling the resulting microstructure.

Algorithm 1: Initial interval temperature estimation

Result: $\hat{T}^j_{0i}(x, t_{i-1})$
$j = 0$;
Calculate $T^H_0(x, t_{i-1}, T_0)$ for $i \in \{1...N\}$;
$\hat{T}^0_{0i}(x, t_{i-1}) = T^H_0(x, t_{i-1}, T_0)$;
while $|\hat{T}^j_{0i} - \hat{T}^{j-1}_{0i}| > tol$ ***and*** $j \leq maximum\ iterations$ **do**
  for $i = 1...N$ **do**
    Calculate $T^H_i(x, t, \hat{T}^j_{0i})$ for $t = t_k, k \in \{i...N\}$;
    Calculate $T_i(x, t, \hat{T}^j_{0i}, Q_i)$ for $t = t_k, k \in \{i...N\}$;
    Calculate $\Delta T_i(x, t, Q_i, \hat{T}^j_{0i}) = T_i(x, t, Q_i, \hat{T}^j_{0i}) - T^H_i(x, t, \hat{T}^j_{0i})$ for $t = t_k, k \in \{i...N\}$;
  end
  $\hat{T}^{j+1}_{0i} = \sum_{k=1}^{i-1} \Delta T_k(x, t_{i-1}, Q_k, \hat{T}^j_{0k}) + T^H_0(x, t_{i-1}, T_0)$;
  $j = j + 1$;
end

Table 4. Parallel-in-time definitions.

$T^H_0(x,t,T_0)$	$\triangleq$ temperature at time t due to homogeneous evolution of the initial temperature T_0
$\widehat{T}^j_{0i}(x,t_{i-1})$	$\triangleq$ estimated initial temperature in iteration j at the start of interval i ($t = t_{i-1}$)
$T^H_i(x,t,\widehat{T}^j_{0i})$	$\triangleq$ temperature at time $t \geq t_{i-1}$ due to homogeneous evolution of initial conditions $\widehat{T}^j_{0i}$
$T_i(x,t,\widehat{T}^j_{0i},Q_i)$	$\triangleq$ temperature at time $t \geq t_{i-1}$ due to heat source Q_i and initial conditions $\widehat{T}^j_{0i}$
$\Delta T_i(x,t,Q_i,\widehat{T}^j_{0i})$	$\triangleq$ $T_i(x,t,Q_i,\widehat{T}^j_{0i}) - T^H_i(x,t,\widehat{T}^j_{0i})$: Change in temperature at time $t \geq t_{i-1}$ due to heat source Q_i

Microstructure (1b). TruchasPBF is a continuum code and, while it gives an accurate representation of the thermal history during solidification, it does not directly model the microstructure formation. As such, we have developed a version of the cellular automata (CA) model of solidification designed for exascale computing (Figure 3), ExaCA. CA was chosen over Monte Carlo (MC) because of the transparent representation of the physics of solidification and over phase field for its low relative computational cost. TruchasPBF simulations at the resolution of a few micrometers are used to drive the ExaCA simulations. The ExaCA output includes a regular grid of grain ID data, where each grain ID is associated with a single grain orientation, that is passed to ExaConstit for modeling of constitutive properties.

ExaCA uses temperature data passed from the CFD models TruchasPBF or OpenFOAM in its calculations of grain structure. A sparse version of the full time-temperature history of the domain is used by the CA, consisting of, for each heat transport node that underwent melting and re-solidification, the time values at which the node cooled below the liquidus and solidus temperatures for the final time. A previous study on overlapping melt pools found that this sparse data for standard AM raster patterns yielded microstructures consistent with those from the full time-temperature history with significantly improved performance (Gandin and Rappaz, 2020). These liquidus and solidus time values must be on a regular grid, and can be provided to ExaCA in one of two ways. The first, as illustrated schematically in Figure 11 and used for the code

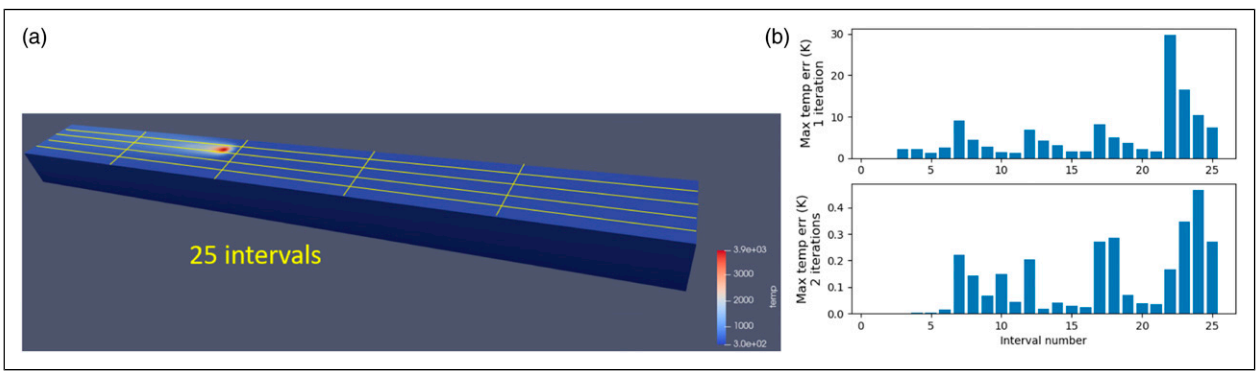

Figure 10. Parallel in time for layer-resolved thermal simulations (Stage 1a): (a) Parallelization of melt pool simulation and (b) time parallelization error.

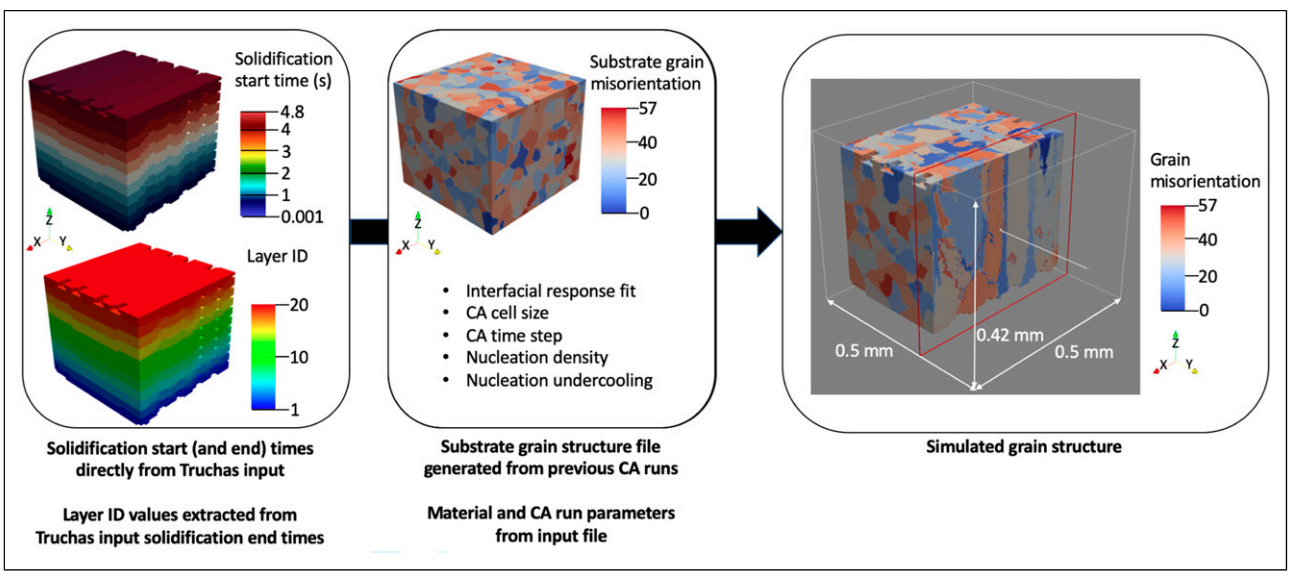

Figure 11. Steps involved in linking multilayer TruchasPBF and ExaCA simulations. TruchasPBF provides liquidus and solidus time data values, which is used along with layer start and end times to extract layer ID values for each CFD node. This is combined with substrate data, information on interpolation from the CFD to the CA length and time scales, and other CA input parameters to simulate a multilayer microstructure.

performance analysis, is used for coupling the model to multilayer TruchasPBF simulations. The time values associated with the start of solidification for each layer are used to extract layer ID values for each data point, that is, the layer number associated with the final time that each TruchasPBF node underwent solidification. The liquidus and solidus time values for each layer are then renormalized using the layer ID values such that each layer begins solidifying at time zero. These normalized liquidus and solidus values are then interpolated from the temperature field data resolution to the CA cell size, for each layer ID. An input file provides material input parameters such as nucleation density and describes how the TruchasPBF data is interpolated to the CA length and time scales. Following this normalization and interpolation, a substrate grain structure is read from a file as an initial condition, and simulation begins.

The second way of handing data to ExaCA involves using temperature data from single layers from OpenFOAM or TruchasPBF. This approach allows a multilayer temperature field to be constructed from the individual layer data, enabling ExaCA to simulate any number of layers with any offset between layers in the build direction and with varied scan patterns (i.e., a unidirectional scan as opposed to alternating even and odd layers). Said data on number of layers, ordering of layers, and build direction offset between layers is taken from an additional ExaCA input file. Once the multilayer temperature field is assembled and layer ID values set for this second initialization approach, the remainder of the workflow mirrors that shown in Figure 11. This second approach is advantageous for the simulation of large numbers of layers independent of additional CFD calculations.

The ExaCA code is based on the decentered octahedron algorithm of Gandin and Rappaz (1997), modified for the sparse version of the full time-temperature history data provided. The as-solidified grain structure for many commonly manufactured alloys (including austenitic stainless steel, Inconel 625 and 718, and β-Ti systems) consists of face-centered cubic (FCC) or body-centered cubic (BCC) crystal structures with six equivalent (100) principal growth directions. Rather than modeling the dendritic grain structure directly, the decentered octahedron method models these grains as octahedral envelopes, where the preferred (001) crystallographic directions are represented by the six octahedron axis directions. The envelopes are tracked in a manner that preserves the orientations of the octahedra on the regular Cartesian grid used by the model. One important aspect of the method is that the growth rate of a given octahedral envelope is not a uniform value, but depends on the local undercooling in the associated CA cell, as well as the interfacial response function for the material. The interfacial response function relates the local undercooling in a CA cell to the growth of the envelope per model time step and is a bridge between the sub-grain physics, which governs dendritic alloy solidification, interfacial kinetics, and solute transport, and part-scale microstructure, which will contain a large number of grains. The velocity-undercooling data needed to construct such a function can come from numerical models of dendrite tip growth, such as the Kurz–Giovanola–Trivedi model (Kurz et al., 1986), other modeling methods, or experimental data. These data are typically fit to a simple function, such as an exponential or a third order polynomial. Despite not modeling the dendritic structure or solute distribution directly, implicitly accounting for these phenomena through use of the interfacial response function and proper crystal geometry has yielded CA-predicted grain sizes, shapes, and textures consistent with AM microstructures observed via EBSD and SEM (Akram et al., 2018; Koepf et al., 2018; Rai et al., 2017; Zinoviev et al., 2016; Zinovieva et al., 2018).

All CA cells associated with the interpolated melting data from TruchasPBF begin as liquid, while the other cells are initialized with grain ID values from the substrate as solid, with the exception of a layer of active cells separating the solid and liquid. Active cells are associated with the solidification front, and have associated octahedral envelopes representative of the local grain orientation. As the octahedral envelopes grow, the decentered octahedron algorithm governs the transformation of adjacent liquid cells into new active cells, which is associated with the creation of new octahedra for the new active cells. Since using the sparse data format implies that each CA cell will undergo solidification once (i.e., remelting is neglected), an active cell with no remaining liquid cell neighbors is transformed into a solid cell and remains solid through the remaining duration of the simulation. Much of this solidification is via epitaxial growth of substrate grains, but heterogeneous nucleation of new grains is modeled as well. This is done by selecting liquid CA cells at random during initialization, with an input density and a distribution of nucleation undercooling values. Any of these cells that remain unclaimed by another growing grain upon cooling to their nucleation undercooling becomes the site of a new grain, and a new octahedral envelope is placed at its center. Algorithm 2 summarizes the CA computational workflow including the initialization, general steps in modeling nucleation and growth, and output.

Algorithm 2: Computational workflow for ExaCA simulation of as-solidified grain structure.

```
Result: ExaCA simulation
Read input files for CA parameters, temperature and substrate data (from TruchasPBF/OpenFOAM);
Interpolate temperature data to CA cell size and initialize fields ;
Initialize CA cell types and initial grain structure ;
Initialize potential nucleation sites and undercooling values ;
for layernumber = 0...N-1 do
    while Cells with melting and solidification data associated with layernumber remain liquid do
        Simulate potential nucleation of new active cells;
        Simulate growth of octahedra in active cells;
        Transform liquid cells adjacent to active cells into new active cells (decentered octahedron algorithm) ;
        Calculate octahedra attributes of new active cells;
        Transform active cells no longer at the solidification front into solid cells ;
        Update cell states and attributes on other processors;
    end
end
Print grain ID data for selected volume of simulation (for ExaConstit);
```

ExaCA is written in C++ and uses Kokkos for on-node performance portability across hardware (CPUs and GPUs) (Edwards et al., 2014) and MPI to parallelize across nodes using either 1D or 2D domain decomposition schemes. The code has been run successfully on Summit for multilayer grain structure calculations using both OpenFOAM and TruchasPBF input data. Ten layers of raster scan patterns alternating direction every other layer and covering various areas are used to test the performance and scaling of ExaCA; the scan areas of these problems are on the order of the small and medium legs of the NIST AM-Bench-01 test part. TruchasPBF data at a 10 μm resolution was interpolated to a 2.5 μm CA cell size, yielding simulation volumes between 18 and 148 million CA cells. Comparing ExaCA performance across hardware, the smallest problem size using a single GPU with the Kokkos CUDA backend was approximately 3× faster than using a single node of CPUs (42 MPI ranks) with the Kokkos OpenMP backend. This difference increases to 5× when comparing six GPU (1 node) to six CPU nodes for the largest problem size.

Strong scaling is defined by how CPU time varies with the number of processors for a fixed total problem size and is used to determine if a method as implemented is compute-bound. Ideal strong scaling is achieved if the CPU time is inversely proportional to the number of processing elements used. Scaling tests for these problems were performed using GPUs on Summit,[5] the results of which are illustrated in Figure 12. For the smallest problem size, little to no speedup (less than 1.2×) was seen in going from a single GPU of Summit to multiple GPUs: there is only enough work to effectively use one GPU. More significant strong scaling

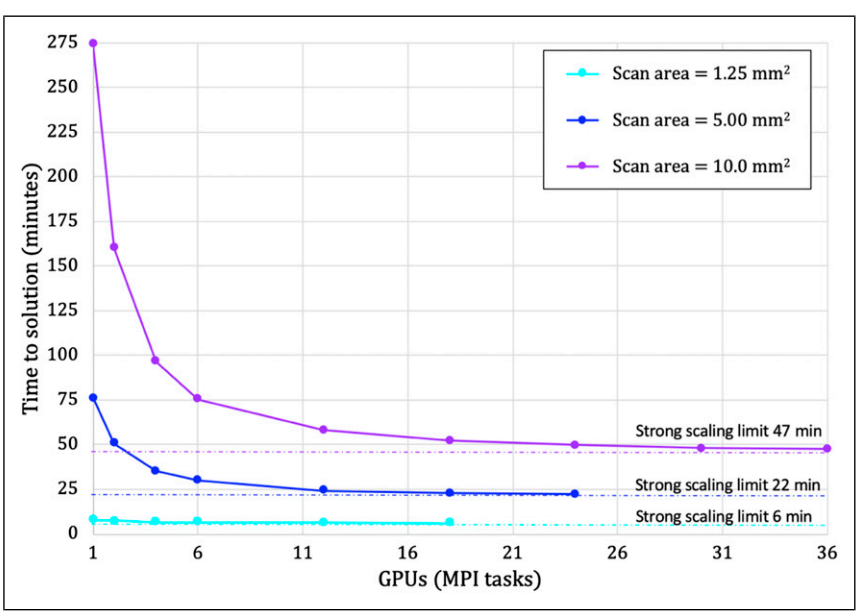

Figure 12. Strong scaling of ExaCA simulation of grain structure on Summit, using 10 layers of temperature data (alternating raster scan pattern) generated by TruchasPBF.

was observed for the intermediate problem size, where scaling up to around 12 GPUs (2 nodes of Summit) results in 3.4× single GPU performance. The largest problem size showed the most significant strong scaling of the three scan areas, up to 24 GPUs (4 nodes of Summit) and 5.8× speedup relative to single GPU performance. The strongly localized nature of solidification calculations along melt pool boundaries, which is small compared to the overall simulation volume, together with the significant cost of MPI communication compared to GPU computation are responsible for the lack of strong scaling. Current development is focused on code performance, for example, optimizing the data layout in memory, and scalability, through minimizing kernel launch overhead, overlapping of communication and computation, and potential sparse grid approaches.

Sub-grain microstructure. While the CA approach used for simulating the microstructure allows for computations sufficient to resolve the grain-scale microstructure, thus allowing modeling of very large physical domains, it does not resolve details within the grains, such as alloy microsegregation and "sub-grain" morphological details. For that, the phase-field model (PFM) (Boettinger et al., 2002; Moelans et al., 2008) is more appropriate and we are modifying our PFM codes for exascale computing to address these issues (Figure 3).

The two main ingredients of a PFM approach are (i) the use of an order parameter field that takes different values in different phases and smoothly varies between the two at the interfaces and (ii) the derivation of time evolution equations based on minimizing an energy functional over time.

The energy functional we use takes the general form

$$F[\varphi,T,c] = \int_V \frac{\varepsilon^2}{2}|\nabla\varphi|^2 + 16\omega\varphi(1-\varphi) + f_{bulk}(\varphi,T,c) \quad (15)$$

where we used the order parameter $\phi = 0$ to denote the liquid phase, and $\phi = 1$ for the solid phase. To simplify the presentation, we denote by c the composition of a binary alloy, but the formulation generalizes to multi-species alloys. The first two terms on the r.h.s. are nonzero only in the interface region, and lead to a solution with a diffuse interface of width $\lambda = \varepsilon/\sqrt{32\omega}$ and energy $\sigma = \varepsilon\sqrt{16\omega}/3\sqrt{2}$. The last term on the r.h.s. is the "bulk" energy density in either the liquid or solid phase, or a mix of both through the interface.

To enable the use of realistic values for the bulk energy, we use the CALPHAD formalism and associated databases. These provide energies for each phase as a function of

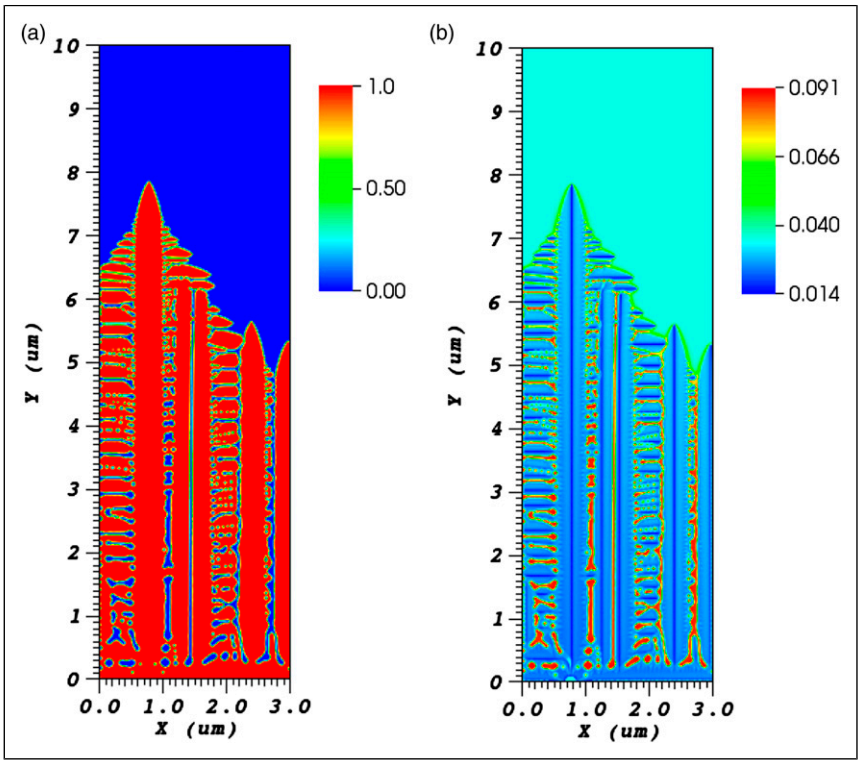

Figure 13. 2D simulation of dendritic growth in a ternary alloy, FeNbNi, after 60 μs for a 3 μm × 10 μm computational domain, starting from a small solid seed at the bottom at time t = 0. This simulation used the CALPHAD database of Mathon et al. 2009: (a) Phase variable and (b) Nb composition.

phase, temperature, and alloy compositions in a simple analytic form with a small number of parameters. To integrate these functions in our formalism, we follow the KKS model proposed by Kim et al. (1999) which assumes coexistence of solid and liquid at each point in space with compositions values associated with each phase, c_S and c_L, that satisfy the system of equations

$$c = (1-\varphi)c_L + \varphi c_l \tag{16}$$

$$\frac{\partial f^S}{\partial c_S} = \frac{\partial f^L}{\partial c_L} \tag{17}$$

where $f^{S,L}$ denote the free energies in the solid (resp. liquid) phases. Using c_S and c_L, one can evaluate

$$f_{bulk}(\varphi,T,c) = \big(1-h_\varphi(\varphi)\big)f^L(c_L,T) + h_\varphi(\varphi)f^S(c_S,T) \tag{18}$$

The time evolution equation for the order parameter ϕ is given by the Allen–Cahn equations

$$\dot{\varphi} = -M\frac{\delta F}{\delta \varphi} \tag{19}$$

where M is the phase-field mobility parameter. The time evolution equations for the composition and temperature fields are diffusion equations

$$\dot{c} = \nabla \cdot D(\varphi)\nabla c \tag{20}$$

$$c_p\dot{T} = \nabla \cdot k\nabla T + L\frac{\partial \varphi}{t} \tag{21}$$

where c_p is the heat capacity per unit volume, k is the thermal conductivity, and L is the latent heat per unit volume. In practice, the temperature diffusion equation is often replaced by a prescribed temperature field to avoid the cost of a reduced time step due to the fast thermal diffusion.

This model has been implemented in the open source code AMPE,[16] using a finite volume approach and an implicit and adaptive time integration. It relies on the software packages SAMRAI[17] for meshing and data distribution, SUNDIALS[18] (Hindmarsh et al., 2005) for the backward Euler time integration, and HYPRE (Falgout et al., 2006) for preconditioning the linear system solves (Structured-Grid System Interface). More details can be found in Dorr et al. (2010) and Fattebert et al. (2014). Figure 13 shows the dendritic structure modeled with AMPE during solidification of a ternary alloy.

In PFM, the diffuse interface between phases is a convenient way to ensure "smooth" numerical representation of the solution on a mesh that just needs to be sufficiently fine to represent that transition. The width of this interface can be made very wide, without affecting numerical solvers, as long as it allows to resolve the finest features of a grain we are interested in resolving. One needs to be careful however to avoid having solutions that depend on that non-physical interface width. For studying alloys solidification in particular, it has been shown that using an interface width too wide can lead to spurious effects and quantitatively affect the results of such simulations (Karma, 2001; Kim, 2007). To avoid affecting quantitative results, simulations parameters should be chosen such that one remains within the "thin interface" limit, that is, $\lambda v/D \ll 1$, where λ is the PFM interface width, v the moving interface velocity, and D is the average diffusivity of the alloy composition through the interface. This obviously poses a challenge for AM applications where v can be of the order of 0.1 m/s and D is of the order of 10^{-9} m^2/s in the liquid phase, leading to PFM

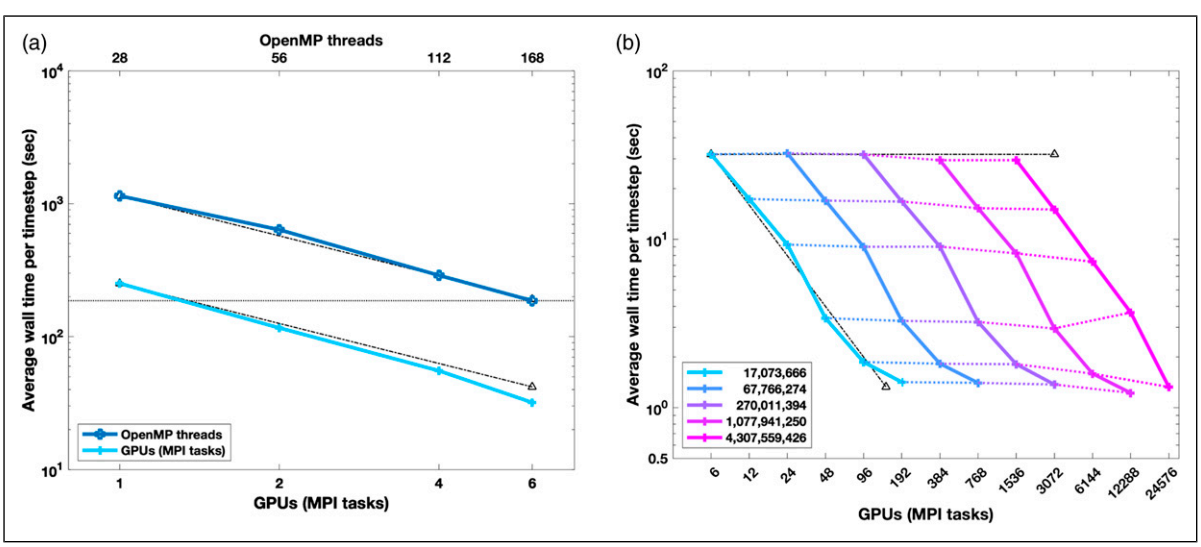

Figure 14. Tusas performance on Summit: (a) Strong and weak single-node, OpenMP and GPU scaling and (b) strong (solid lines) and weak (dotted lines) multi-node, multi-GPU scaling.

interface width of the order of 1 nm, and even smaller mesh spacing.

At the time of this writing, while AMPE implements most of the physics models needed for modeling solidification in AM conditions, it does not have the capability of using GPU accelerators. So in order to evaluate performance of PFM on pre-exascale platforms, we use the Tusas code.

Tusas is a general, flexible code for solving coupled systems of nonlinear partial differential equations. Tusas was originally developed for phase-field simulation of solidification. The Tusas approach consists of an unstructured Lagrange finite element spatial discretization of the fully coupled nonlinear system, which is treated explicitly or implicitly (Euler, Trapezoid, BDF2, or IMR) in time with a preconditioned Jacobian-free Newton–Krylov (JFNK) method. The preconditioning strategy in Tusas is based on block factorization and algebraic multigrid that allows an efficient, implicit time integration. As the JFNK method only requires a residual, from an implementation standpoint, Tusas allows a flexible framework as it only requires the user to implement code for the residual equation; with configuration of the nonlinear system and preconditioner performed at run time.

In addition to performing well in parallel across multiple CPUs using MPI and OpenMP, a particular strength of Tusas (Trainer et al., 2016; Ghosh et al., 2020) is the ability to run efficiently across multiple GPUs using MPI and CUDA. As the residual fill is composed entirely of local operations, it is performed entirely on the GPU utilizing a distance-1 element graph coloring within each MPI subdomain. Element graph coloring avoids shared memory race conditions and atomic operations on CPU threads and GPUs. In addition, inner products within the GMRES solver are implemented on threads and each GPU via OpenMP and CUDA within the Belos and Kokkos packages in Trilinos.[19]

To demonstrate strong and weak parallel multi-node and multi-GPU scaling, we consider simulations for a 3D dilute binary alloy solidification similar to Farzadi et al. (2008), with discretizations consisting of 17,073,666; 67,766,274; 270,011,394; 1,077,941,250, and 4,307,559,426 unknowns. The discretizations consist of meshes with $512 \times 128 \times 128$, $512 \times 256 \times 256$, $512 \times 512 \times 512$, $512 \times 1024 \times 1024$, and $512 \times 2048 \times 2048$ bilinear hexahedral elements.

Whereas strong scaling was defined earlier, weak scaling is defined as how CPU time varies with the number of processors for a fixed problem size per processor and is used to determine if a method as implemented is memory bound. Ideal weak scaling is achieved if the CPU time remains constant while the workload is increased in direct proportion to the number of processors.

Figure 14(a) shows the CPU time required for time integration to a fixed time on a single node of Summit[5] as a function of the number of GPUs and number of OpenMP threads with 17,073,666 unknowns. Ideal strong scaling is demonstrated for OpenMP threads with up to 168 threads, and for up to six GPUs. A speedup of approximately 6× is achieved by utilizing six GPUs over 168 threads. Figure 14(b) shows the CPU time required for time integration to a fixed time as a function of the number of GPUs across multiple nodes, utilizing six GPUs per node on Summit,[5] and Figure 14(b) shows strong and weak scaling up to 24,576 GPUs (4096 nodes) with up to 4,307,559,426 unknowns. Strong scaling is depicted by solid lines on Figure 14(b), where each color depicts a fixed problem size. Weak scaling is depicted by horizontal markers with dashed lines. We acknowledge that further optimizations can be performed on our GPU implementation. These preliminary scaling studies demonstrate that Tusas effectively scales strongly and weakly on thousands of GPUs with billions of unknowns and is particularly suited for emerging heterogeneous architectures.

Stage 2: Late-time microstructure

In ExaAM, we use the phase-field method as implemented in MEUMAPPS-SS (Microstructure Evolution Using Massively Parallel Phase-field Simulations—Solid State) (Radhakrishnan et al., 2016) to predict the "late-time microstructure" (Figures 3 and 4). Because the physics of late-time evolution in the solid state is driven by diffusion kinetics in the presence of a stress field (not present in the liquid state), a Fourier spectral approach was taken, which ties the performance of MEUMAPPS-SS on exascale platforms to that of exascale FFT solvers.

Post-build heat treatments are often necessary for parts produced during AM of structural alloys because the as-solidified microstructure is often inadequate for meeting the desired property targets (Kok et al., 2018). In solution hardened alloys such as IN625, solute gradients arising from microsegregation have to be homogenized. In precipitation hardened alloys such as IN718 (hardened by the γ'' phase) and Haynes 282 (hardened by the γ' phase), the cooling rates subsequent to solidification are generally too high in laser powder bed fusion processes to permit diffusion-based precipitation reactions. Therefore, a heat treatment is necessary to precipitate the strengthening phase(s). In contrast, precipitation can occur during electron beam powder-based fusion processes where the heated base plate lowers the post-solidification cooling rate. Here, we are mostly concerned with precipitation during a post-build heat treatment because the challenge problem involves a laser powder bed fusion process.

Phase-field models are a leading approach for precipitation simulations because the evolution of the smoothly varying order parameters describes the motion of the phase boundaries without explicitly tracking the interface and

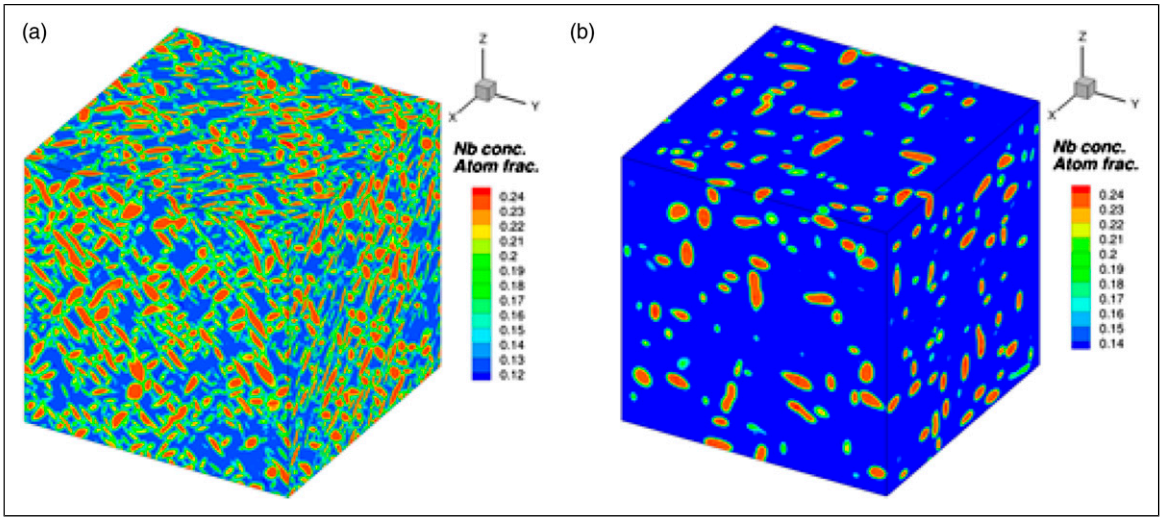

Figure 15. MEUMAPPS-SS simulations of precipitation of γ'' and δ phase in a Ni-Fe-Nb alloy. Higher interfacial energy suppresses δ formation and promotes γ''. Performed on 252^3 domain with 1764 CPUs and 252 GPUs on Summit: (a) $\gamma''-\delta$ interfacial energy = 0.2 J/M^2 and (b) $\gamma''-\delta$ interfacial energy = 0.4 J/M^2.

because the governing equations are directly tied to thermodynamics of the system (Moelans et al., 2008; Steinbach and Apel, 2006). Unlike computationally inexpensive mean-field models for precipitation such as the KWN model (Wagner and Kampmann, 1991) and its variants, phase-field models can capture the detailed interactions between precipitates and allow arbitrary precipitate shapes (Radhakrishnan et al., 2016; DeWitt and Thornton, 2018). MEUMAPPS-SS uses an extended version of the Kim-Kim-Suzuki model (originally developed for a binary alloy (Kim et al., 1999)) to simulate the evolution of an arbitrary number of phases in an alloy with an arbitrary number of components (Zhou, 2014). This model involves the solution of a coupled system of time-dependent Ginzburg–Landau equations, Cahn–Hilliard equations, and mechanical equilibrium with linear elasticity using the well-known Fourier spectral method with semi-implicit time integration (Che and Shen, 1998).

One of the goals of MEUMAPPS-SS in ExaAM is to take "as-built" microstructures such as those generated in Stage 1 of the ExaAM workflow and simulate the microstructural evolution in the solid state either during the build process itself or during post-process heat treatment. Some of the complexities associated with the AM solidification structures include (1) presence of dislocations of unknown density, which has a direct bearing on the density of nucleation sites for solid state precipitation, (2) steep gradients in solute concentration, and (3) uncertainties in parameters that influence the nucleation and growth of precipitates including interfacial energies and their anisotropy, transformation strains, and elastic properties of the matrix and the precipitate. The goal of the phase-field modeling effort is to perform an uncertainty analysis based on ranges in the input parameters to the model and to understand why a set of parameters giving a close match between the simulations and experimental measurements is physically justified under AM conditions. Example simulations in Figure 15 show that when the interfacial energy between the δ phase and the matrix is increased, the preferred precipitation changes from δ to γ''. The length scale of MEUMAPPS-SS simulations is controlled by the length scale of the features found in the solidification structure. In order to account for intra- and intergranular evolution of microstructure, the simulation domain should comprise sufficient number of grains/dendrites in the solidification structure. Based on observed grain sizes in the AM-Bench experiments performed on IN625, the domain edge should be on the order of 10 μm. The length scale of the precipitates are in the nanometer range, which determines the mesh resolution. It is estimated that simulation domain sizes of the order of 2000 × 2000 × 2000 mesh points are required to capture the statistical features present in the solidification microstructure. Solving such problems for the typical times involved in post-process anneals will require exascale capabilities.

The original version of MEUMAPPS-SS is written in Fortran 90 and uses a parallel FFT package, P3DFFT (Pekurovsky, 2012), that allows MPI decomposition of the computational domain in two directions (a "pencil decomposition"), which, in turn, enables the use of scalable 3D FFTs, where the 1D FFTs are performed with the FFTW library (Frigo and Johnson, 2005). This version exhibited near-ideal scaling up to 4096 CPU cores on the OLCF Titan computer for a calculation with 20 nm grid size (Radhakrishnan et al., 2016). OpenACC directives are used to leverage GPUs on current pre-exascale systems and to

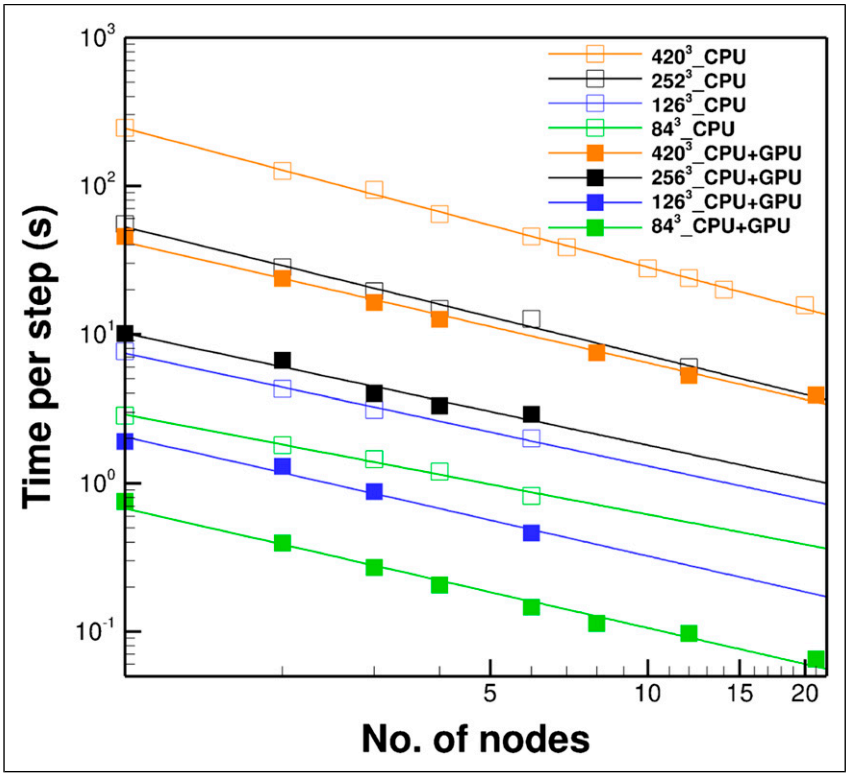

Figure 16. MEUMAPPS-SS scaling on Summit.

prepare for future exascale platforms. The scaling of the Fortran 90 version of MEUMAPPS-SS on Summit[5] with or without GPU off-load using OpenACC is shown in Figure 16. The replacement of OpenACC with OpenMP off-loading and of P3DFFT with heFFTe (a GPU-enabled scalable 3D FFT library (Ayala et al., 2020)) is under active investigation, as is use of the ADIOS 2 library (Godoy et al., 2020) for output and checkpoint/restart functionality.

Development of a more modern and modular version of MEUMAPPS-SS, written in C++, is also underway. This version uses Kokkos (Edwards et al., 2014) for performance portable node-level parallelization. Current development is focused on use of the CUDA backend for NVIDIA GPUs and the HIP backend for AMD GPUs, with planned support for other architectures as they become available. Calculations with the C++ version of MEUMAPPS-SS can perform GPU-enabled 3D FFTs using either the heFFTe library (Ayala et al., 2020) (CUDA and HIP backends, support for multiple nodes) and the AccFFT library (Gholami et al., 2015) (CUDA backend, currently limited to single-node support). Both libraries use pencil MPI domain decompositions and use the FFTW library for 1D FFTs on CPUs and a vendor library (e.g., cuFFT[20] and rocFFT[21]) for 1D FFTs on GPUs. Integration with both ADIOS 2 for I/O (Godoy et al., 2020) and SUNDIALS (Hindmarsh et al., 2005) for time integration is under investigation. Many of the libraries, programming models, and runtimes discussed above are part of the ECP software ecosystem (i.e., MPI, OpenACC, OpenMP, heFFTe, ADIOS 2, Kokkos, and SUNDIALS), which allows MEUMAPPS-SS to leverage their development for both portability and performance improvements for exascale platforms such as Frontier and Aurora.

To test the GPU speedup of MEUMAPPS-SS simulations, a series of simulations were performed of precipitate evolution in an IN625 surrogate involving three variants of the γ'' phase and 12 variants of the δ phase. These simulations had a 168^3 grid and were performed on one node of Summit.[5] The Fortran/OpenACC version of MEUMAPPS-SS exhibited a 9× nodal GPU speedup and the C++/Kokkos version exhibited a 5× nodal GPU speedup (where the CPU-only baselines used all 42 CPU cores). Despite the lower nodal GPU speedup, the run time for the fastest single-node C++/Kokkos calculation 7× lower than that of the fastest single-node Fortran/OpenACC calculation (and 57× lower than the CPU-only Fortran calculation), likely due to a combination of the use of a GPU-accelerated FFT library, the effectiveness of Kokkos for non-FFT portions of the code, and improved memory management. For this single-node simulation, the C++/Kokkos version spent 81% of the run time performing the 3D FFT calculations. This percentage increases to nearly 100% for multi-node calculations because almost all MPI communication necessary in a

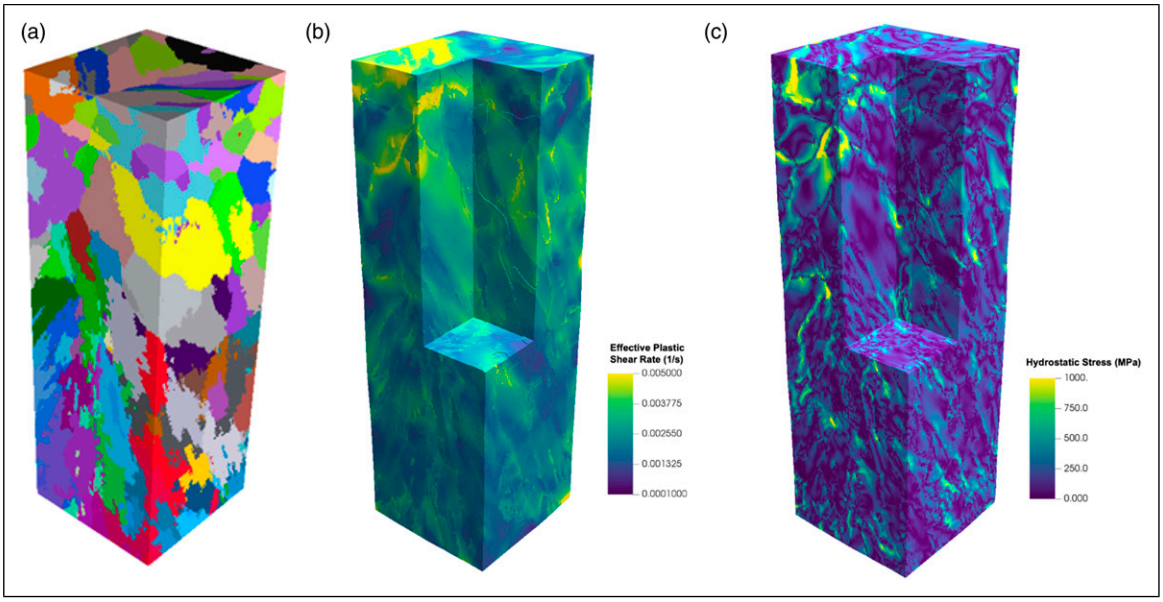

Figure 17. ExaConstit simulations of a monotonic tension test out to 10% strain for a stainless steel 316L microstructure generated by ExaCA: (a) Microstructure generated by ExaCA, (b) plastic strain rate, and (c) hydrostatic stress.

typical time step is contained in the FFT library calls. These observations highlight the importance of high-performance FFT libraries to this portion of the project.

Stage 3: Micromechanical properties

Running full-scale crystal plasticity finite element (CPFEM) simulations at each quadrature point within macroscale simulations, FE^2 methods (Feyel, 2003), is still outside the realm of routine calculations. Depending on the problem, these FE^2 methods are not appropriate when the microstructure starts approaching the size of the macroscopic part (Yvonnet, 2019). However, previous work (Barlat et al., 2005; Barton et al., 2008; Gawad et al., 2015; Knap et al., 2008; Segurado et al., 2012; Van Houtte et al., 2009; Zhang et al., 2016) has devised a number of different ways to use crystal plasticity simulations to inform the macroscale simulation. Rather than using less accurate and less computationally expensive homogenization methods for the crystal plasticity simulations, CPFEM was chosen for two general reasons. First, it provides a higher fidelity in the stress-strain response. Second, CPFEM is capable of properly modeling the intra-grain heterogeneous deformation response as seen in Figure 17 where distributions of the hydrostatic stress and effective plastic shear rate are shown for a monotonic tension test of an ExaCA generated microstructure of a stainless steel 316L alloy. This heterogeneity is of interest in aspects of the PSPP relationships mentioned in the introduction, for example, in driving porosity models.

The ExaAM workflow (Figures 3 and 4) connecting the microstructural-scale response to the part-scale response will not diverge from the already established micro-to-macro homogenization methods. The micromechanical response will be used to inform macroscopic yield surface models and macroscopic porosity models for a given representative volume element (RVE). However, one of the principal differences in the workflow here is the use of a microstructure constructed from the underlying physics of the problem. Also, the approach is built around a workflow suited for exascale platforms, Frontier and El Capitan, enabling rapid turnaround and high-accuracy within the ExaAM workflow. This workflow is necessary to run the significant number of simulations needed for homogenization using meshes with millions of elements that capture the AM-produced microstructures.

Even with advances in computational approaches and platforms, resolving the 3D aspects of AM microstructures is computationally intensive, and the approach employed in calculations here is tailored to the overall workflow. Rather than introducing the complexities and expense of grain-boundary-conforming finite element meshes, we use a rectilinear mesh. Based on previous work (Chatterjee et al., 2019), this is not a major source of error given the small strains and homogenized properties of interest here. In order to accomplish these goals, the CPFEM code-base needs to be designed to take full advantage of the hardware found in current and future supercomputers such as Summit/Sierra and Frontier/El Capitan.

ExaAM makes use of three open source code-bases for these CPFEM simulations, two of which (ExaCMech and ExaConstit) are supported by ExaAM. The crystal plasticity constitutive response is handled by the library ExaCMech

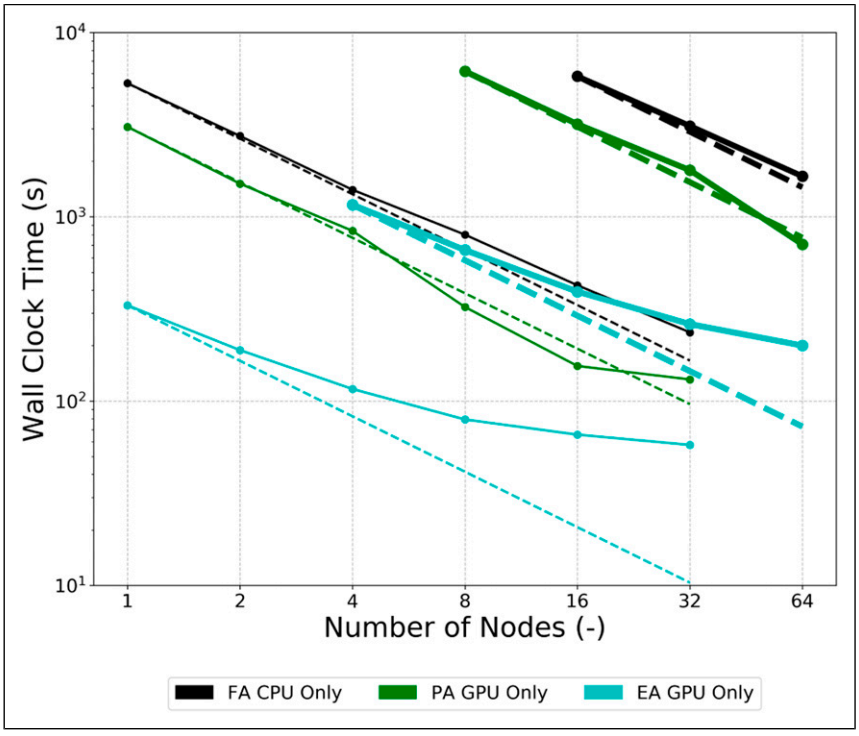

Figure 18. Strong multi-node, multi-GPU scaling on Summit for various assembly methods. Thinner lines represent the 1 M element mesh, and the thicker lines represent the 8 M element mesh. Dashed lines represent perfect scaling for each assembly method and mesh size.

(Barton et al., 2018). It is coupled with ExaConstit (Carson et al., 2019), a general quasi-static nonlinear solid mechanics velocity-based finite element application built on the MFEM framework.[22] ExaConstit solves the linear momentum portion of equations (3). ExaCMech makes use of the RAJA library (Hornung and Keasler, 2014) to abstract away the CUDA and HIP-type kernel launches. RAJA is used to wrap the entire constitutive model into one large compute kernel that is launched on the device. Within ExaConstit, different GPU strategies are being examined. The dominant computational cost within ExaConstit is the series of linearized system solves within a Newton–Raphson scheme to obtain the velocity field. In the currently employed approach, the linearized system matrix is never fully assembled on the GPU. Instead, matrix-free methods are used which perform the action of the stiffness matrix (Anderson et al., 2020; Gupta, 1983; Gupta and Mohraz, 1972). A disadvantage of using matrix-free methods is the loss of traditional preconditioners that are used within most FEM implementations. Therefore, ExaConstit resorts to a matrix-free Jacobi preconditioner. Depending on the assembly method—partial assembly (PA) on the GPU (Anderson et al., 2020; Gupta and Mohraz, 1972), element assembly (EA) on the GPU (Gupta, 1983), or full assembly (FA) on the CPU—ExaConstit exhibits different strong scalability on Summit. This is shown in Figure 18 for meshes with 1 and 8 million linear hexahedron elements, in simulations where the polycrystal undergoes monotonic loading out to 1% strain. The PA method exhibits perfect scaling, FA exhibits acceptable scaling, and the EA exhibits poor scaling. Any GPU-only simulation used six GPUs and six CPUs per node, and the CPU-only simulations made use of all CPUs on a given node. While the EA method does exhibit poor scaling, it is by far the most performant and allows us to run the hundreds of simulations needed for the challenge problem all concurrently. Further speed-ups, as well as reduced start-up costs, could be obtained by using more efficient linear and nonlinear solvers.

Stage 4: Full part build with process-informed, locally accurate properties

With the spatially varying thermal history, associated predictions of microstructural features, and suitable constitutive homogenization procedures in hand, Stage 4 of the ExaAM workflow (Figures 3 and 4) begins by updating the macroscale constitutive model description to be consistent with the AM build conditions. An updated part-scale build simulation is performed to refine predictions of build features such as residual stress and porosity development.

The AM microstructure and texture shown in Figure 17 is reminiscent of sheet metal processing. With this observation in mind, we borrow on decades of research modeling anisotropic material response, and start with what's classically referred to as Barlat's Yld2004-18p yield surface model (Barlat et al., 2005). This highly adaptable model reduces to either the classical anisotropic Hill or J2 plasticity models as the data indicates. The largely orthotropic model has 14 parameters determined by in-plane properties and the last 4 determined by out of plane properties. We anticipate the need to extend the model as we build our database of AM microstructures and their material response.

Algorithm 3 shows the computational workflow for determining an optimized AM process-aware constitutive model from our simulated AM microstructures. The parameters in the macroscale constitutive model depend nonlinearly on microstructural descriptors, such as the grain size and heterogeneous alloy distributions. Many direct numerical simulations using ExaConstit are needed to homogenize over these microstructrural descriptors and to probe anisotropy in the flow stress response. These independent simulations lead to a natural utilization model for the exascale computer: an ensemble workflow which assimilates an optimized macroscopic model from microscale simulations. Feeding this AM process-aware constitutive model back into Stage 0 enables optimization of process parameters for residual stress, distortion, and location-specific material properties, as well as assessment of the overall performance of the part.

Algorithm 3: Computational workflow to determine continuum AM process-aware constitutive model.

```
Result: AM process-aware constitutive model
Identify locations with distinct AM process conditions;
for each location do
    Generate RVE;
    Read ExaCA output file;
    Output grain ID and orientation for each voxel;
    Generate MFEM mesh;
    for ensemble of temperatures and loading directions do
        Run ExaConstit;
        Generate data for constitute model;
    end
end
Optimize constitutive model;
Feedback to Stage 0;
```

Summary and next steps

The goal of the Exascale Additive Manufacturing Application Development Project (ExaAM) is to implement an exascale framework for modeling metal AM processing. The multi-physics components of this framework are driven by the realization that exascale capabilities will enable AM process simulation at the fidelity of the microstructure. Each of these component simulation codes is designed to utilize exascale computing. Our computational workflow and exascale utilization model are driven by the coupling between AM processing, material microstructure, material properties, and part performance (PSPP), with the goal of creating AM process-aware material models. In each section, we present details of the information flow between different physics components in our computational workflow. The use of these models within AM build simulations and an Integrated Computational Engineering framework will enable parts that are designed to be manufactured by AM processing, designed with location-specific properties, and are designed within tighter qualification specifications.

In the remainder of the project, we will be developing algorithms and software for high-resolution simulations of the laser–powder interaction and the subsequent melting and solidification to generate improved constitutive models in coarser length scale TruchasPBF calculations where the powder is not resolved. Resolving the powder geometry, the free surface flow of the melt pool, and the resulting voids and solidification will provide effective material properties for both heat transfer and mechanics and models which will better approximate the problem geometry, the resulting thermal gradients and solid-liquid interface velocities, and the resulting residual stresses near the melt pool immediately after solidification in our part-scale simulations. These high-resolution results will also be directly coupled with our CA solidification models to create both a better understanding of local microstructure generation as well as improve the general stage 1 physics coupling procedure under less-resolved geometric conditions. We have already started this process and are developing new models using the material point method (MPM), a variant of the particle-in-cell (PIC) method well suited for problems with dynamic interfaces and large deformations (Sulsky et al., 1994). Our models are designed to be scalable on all DOE exascale platforms (Wang et al., 2020) and build on particle tools from the ECP CoPA co-design center (Slattery et al., 2018). We are validating these models by performing direct numerical simulations of new experiments performed at the Advanced Photon Source (APS) which yield high spatial and temporal resolution images of the LPBF process (Parab et al., 2018).

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Notes

1. Note that this refers to failures in the AM build process, not failure of successfully manufactured parts during use.
2. https://www.exascaleproject.org/
3. https://www.olcf.ornl.gov/frontier/
4. https://www.olcf.ornl.gov/
5. https://www.olcf.ornl.gov/summit/
6. https://hpc.llnl.gov/hardware/platforms/sierra/
7. https://www.llnl.gov/
8. https://en.wikipedia.org/wiki/DDR4_SDRAM
9. https://en.wikipedia.org/wiki/High_Bandwidth_Memory
10. https://en.wikipedia.org/wiki/InfiniBand
11. http://www.nist.gov/ambench
12. http://www.nist.gov/ambench/amb2018-01-description
13. Unified Modeling Language, https://www.uml.org/
14. Note that while these methods do indeed provide performance improvements, they are really more than that. They are *facilitators*, in the sense that a fully resolved thermal-solids solution of the part-scale problem would otherwise be untenable.
15. There exist allusions in the literature to the composition of these methods, and, indeed, the authors have already implemented some of these hybrid methods.
16. http:://github.com/LLNL/AMPE
17. https://computing.llnl.gov/projects/samrai
18. https://computing.llnl.gov/projects/sundials
19. https://trilinos.github.io
20. https://docs.nvidia.com/cuda/cuda-toolkit-release-notes/
21. https://github.com/ROCmSoftwarePlatform/rocFFT
22. http://mfem.org/
23. http://energy.gov/downloads/doe-public-access-plan

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